

ASUS ZS590KS

RF Hardware Trouble Shooting Guide

ZS590KS Supported Bands

2G:	GSM	GSM850/EGSM/DCS/PCS
3G:	WCDMA	B1/B2/B3/B4/B5(B6/B19)/B8
4G:	LTE-FDD	B1/B2/B3/B4/B5/B7/B8/B12/B17/ B18/B19/B20/B26/B28(WW) B25/B30/B66/B71(US)
	LTE-TDD	B34/B38/B39/B40/B41/B42
	4*4MIMO	B1/B3/B7
5G:	NR-FDD	N1/N2/N3/N5/N7/N8/N12/N20/ N28(WW)
	NR-TDD	N38/N77/N78
	4*4MIMO	N1/N3/N7/N77/N78

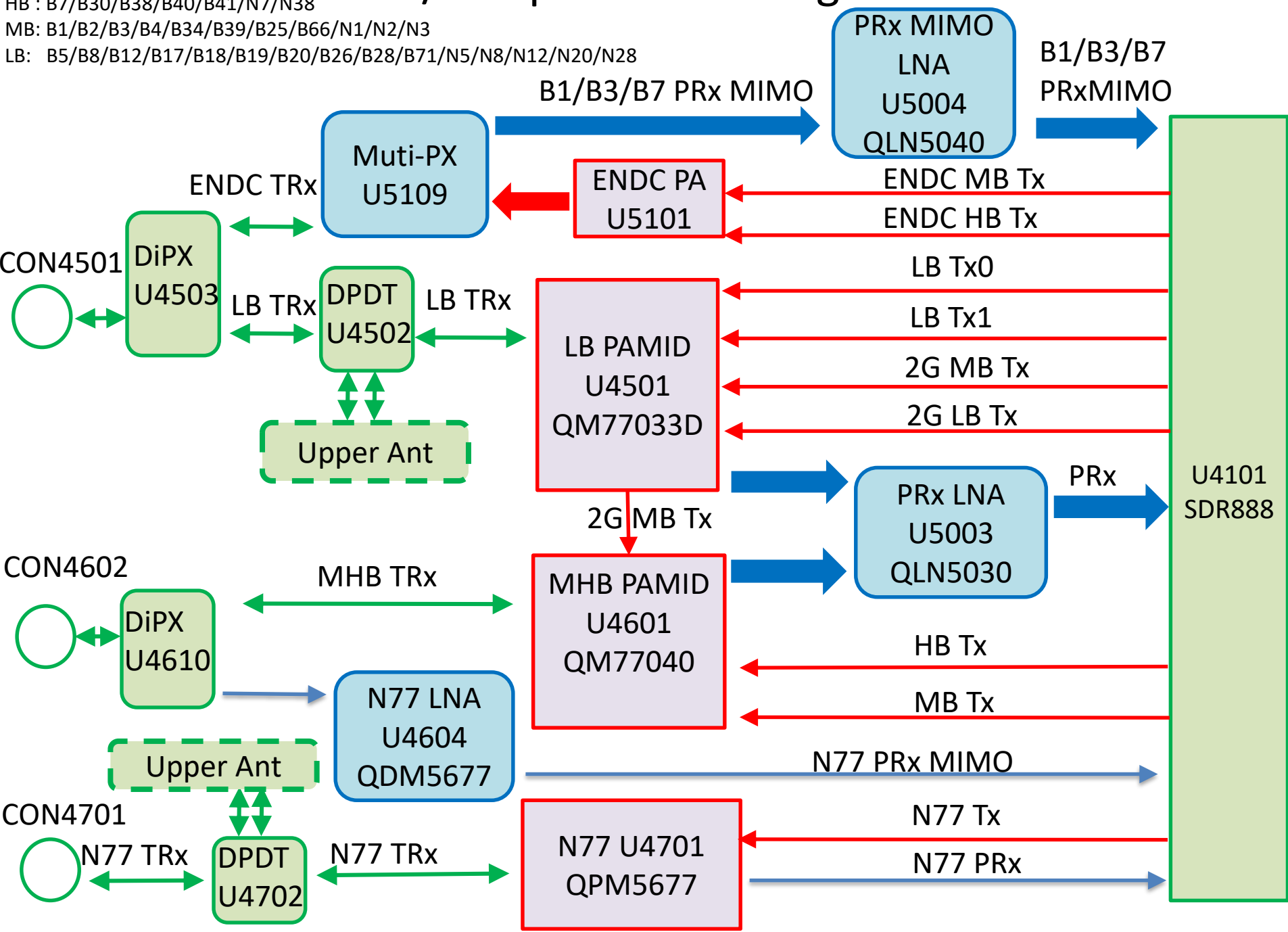
N77: B42/N77/N78

HB : B7/B30/B38/B40/B41/N7/N38

MB: B1/B2/B3/B4/B34/B39/B25/B66/N1/N2/N3

LB: B5/B8/B12/B17/B18/B19/B20/B26/B28/B71/N5/N8/N12/N20/N28

Tx/PRx path block diagram



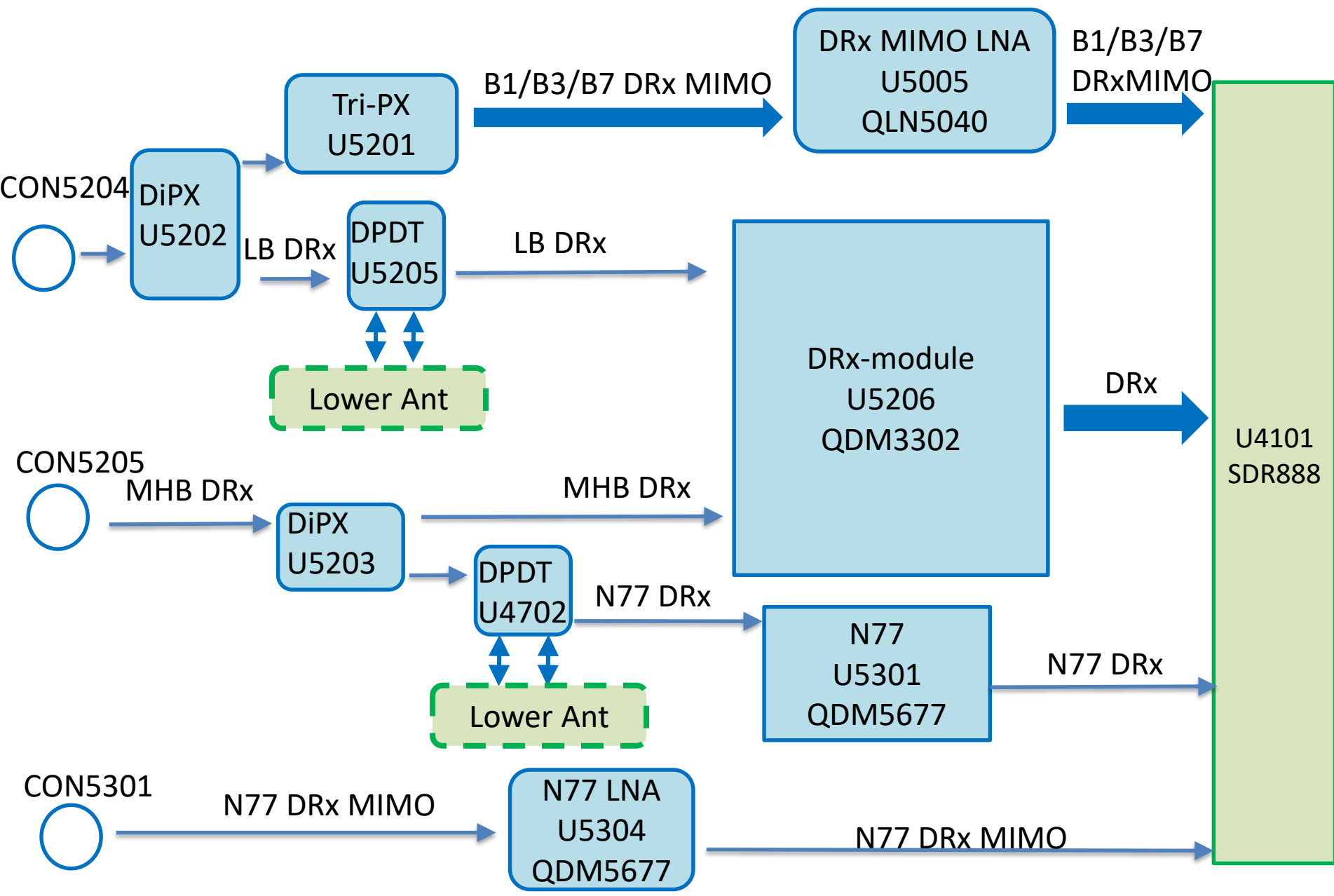
N77: B42/N77/N78

HB : B7/B30/B38/B40/B41/N7/N38

MB: B1/B2/B3/B4/B34/B39/B25/B66/N1/N2/N3

LB: B5/B8/B12/B17/B18/B19/B20/B26/B28/B71/N5/N8/N12/N20/N28

DRx path block diagram



RFFE SCAN fail

Error Code : 3921



1. Computer search
"QFDT" and open

Qualcom Factory Debug Tool

File Tools Help

Product: **SampleProduct** COM: **59** **Connect** Operating Mode: **ONLINE**

Device Information

HLOS/FFBM: **FFBM** FTM/ONLINE NV: **FFBM** **Reset** FTM/ONLINE RUNTIME: **FFBM** Power Supply: **On** **Off**

4

Modem Tx

Modem Rx

GSM

GNSS

WLAN

Factory Info

RFFE Scan

NV Items

HLOS Commands

Shop Floor Control

Power Supply

FTM AP Commands

QMSL Log

5

Report

File Size = 1480
File Version = 2
RFFW ID = 802
Number of Device Instances = 33
Number of Device Instances Present = 33
Error Code = 0

Dev Inst	BUS	MID	PID	P_REV	USID	Status
0	, 0	, (0x0)	, (0x0)	, 0x0	, 0xFF	PRESENT
1	, 32	, (0x0)	, (0x0)	, 0x0	, 0xFF	PRESENT
2	, 32	, (0x0)	, (0x0)	, 0x0	, 0xFF	PRESENT
3	, 32	, (0x0)	, (0x0)	, 0x0	, 0xFF	PRESENT
4	, 32	, (0x0)	, (0x0)	, 0x0	, 0xFF	PRESENT
5	, 35	, QUALCOMM(0x217)	, (0x1AC)	, 0xD0	, 0x8	PRESENT
6	, 37	, QUALCOMM(0x217)	, (0x253)	, 0x10	, 0x3	PRESENT
7	, 35	, QUALCOMM(0x217)	, (0x338)	, 0x0	, 0xD	PRESENT
8	, 35	, QUALCOMM(0x217)	, (0x337)	, 0x1	, 0x4	PRESENT
9	, 37	, QUALCOMM(0x217)	, (0x257)	, 0x0	, 0xB	PRESENT
10	, 39	, QUALCOMM(0x217)	, (0x257)	, 0x0	, 0x8	PRESENT
11	, 37	, QUALCOMM(0x217)	, (0x26E)	, 0x2	, 0xA	PRESENT
12	, 39	, QUALCOMM(0x217)	, (0x26E)	, 0x2	, 0x2	PRESENT

Status PRESENT means pass, MISSING FATAL means no devices.
Please reference dev(devices) location on the next page

Debug Messages

Clear

Error Code : 3921

Device location (WW)

Dev		location					
Device_ID	Component_ID	Bus	MID	PID	Rev	Default_USID	Assigned_USID
[1]	QET5100	U4801A	1	0x217	0x35	0x0	0x4
[2]	QET5100	U4802	2	0x217	0x35	0x0	0x4
[3]	QET6100	U4901	3	0x217	0x337	0x1	0x4
[4]	QM77040	U4601	1	0x134	0x42	0x0	0xF
[5]	QM77033D	U4501	2	0x134	0x45	0x0	0x8
[6]	QM77033D	U4501	2	0x134	0x44	0x0	0xF
[7]	QLN5030	U5003	5	0x217	0x0257	0x0	0x8
[8]	QLN5040	U5004	5	0x217	0x026E	0x2	0xA
[9]	QLN5040	U5005	7	0x217	0x026E	0x2	0xA
[10]	QDM3302	U5213	7	0x217	0x41D	0x1	0xC
[11]	QDM3302	U5206	7	0x217	0x810	0x1	0x8
[12]	QPM5677	U4701	3	0x217	0x01AC	0xD2	0x8
[13]	QPM5677	U4701	5	0x217	0x0253	0x10	0x3
[14]	QPM5677	U4701	3	0x217	0x338	0x0	0xD
[15]	QDM5677	U5301	7	0x217	0x253	0x2	0x4
[16]	QDM5677	U4604	5	0x217	0x253	0x2	0x9
[17]	QDM5677	U5304	7	0x217	0x253	0x2	0x9
[18]							
[19]							
[20]	QM11022	U4502					
[21]	QM11022	U4702					
[22]	QAT3555	U5202	1	0x217	0x687	0x8	0x9
[23]	FX5773STR1	U5007					
[24]	FX5773STR1	U5001					
[25]	RF5426	U5101	2	0x134	0xFD	0x0	0xF
[26]	FX5773STR1	U4606					
[27]	FX5773STR1	U4611					

Error Code : 3921

Device location (US)

Dev	location						
Device_ID		Component_ID	Bus	MID	PID	Rev	USID
[1]	QET5100	U4801A	1	0x217	0x35	0x0	0x4
[2]	QET5100	U4802	2	0x217	0x35	0x0	0x4
[3]	QET6100	U4901	3	0x217	0x337	0x1	0x4
[4]	QM77038	U4601	1	0x3C6	0x010	0x0	0xE
[5]	QM77033D	U4501	2	0x134	0x45	0x0	0x8
[6]	QM77033D	U4501	2	0x134	0x44	0x0	0xF
[7]	QLN5030	U5003	5	0x217	0x0257	0x0	0x8
[8]	QLN5040	U5004	5	0x217	0x026E	0x2	0xA
[9]	QLN5040	U5005	7	0x217	0x026E	0x2	0xA
[10]	QDM3302	U5213	7	0x217	0x41D	0x1	0xC
[11]	QDM3302	U5206	7	0x217	0x810	0x1	0x8
[12]	QPM5677	U4701	3	0x217	0x01AC	0xD2	0x8
[13]	QPM5677	U4701	5	0x217	0x0253	0x10	0x3
[14]	QPM5677	U4701	3	0x217	0x338	0x0	0xD
[15]	QDM5677	U5301	7	0x217	0x253	0x2	0x4
[16]	QDM5677	U4604	5	0x217	0x253	0x2	0x9
[17]	QDM5677	U5304	7	0x217	0x253	0x2	0x9
[18]							
[19]							
[20]	QM11022	U4502					
[21]	QM11022	U4702					
[22]	QAT3555	U5202	1	0x217	0x687	0x8	0x9
[23]	FX5773STR1	U5007					
[24]	FX5773STR1	U5001					
[25]	RF5426	U5101	2	0x134	0xFD	0x0	0xF
[26]	FX5773STR1	U4606					
[27]	FX5773STR1	U4611					

Cal log示意圖

LTE_B1_Connected_V2_FBRX_PRX_TX0	PASS	1.01
LTE_B2_Connected_V2_FBRX_PRX_TX0	PASS	0.59
LTE_B3_Connected_V2_FBRX_PRX_TX0	PASS	1.00
LTE_B4_Connected_V2_FBRX_PRX_TX0	PASS	0.91
LTE_B5_Connected_V2_FBRX_PRX_TX0	PASS	0.50
LTE_B7_Connected_V2_FBRX_PRX_TX0	PASS	1.42
LTE_B8_Connected_V2_FBRX_PRX_TX0	PASS	0.39
LTE_B12_Connected_V2_FBRX_PRX_TX0	PASS	0.43
LTE_B17_Connected_V2_FBRX_PRX_TX0	PASS	0.38
LTE_B18_Connected_V2_FBRX_PRX_TX0	PASS	0.38
LTE_B19_Connected_V2_FBRX_PRX_TX0	PASS	0.38
LTE_B20_Connected_V2_FBRX_PRX_TX0	PASS	0.38
LTE_B26_Connected_V2_FBRX_PRX_TX0	PASS	0.38
LTE_B28A_Connected_V2_FBRX_PRX_TX0	PASS	0.57
LTE_B28B_Connected_V2_FBRX_PRX_TX0	PASS	0.38
LTE_B34_Connected_V2_FBRX_PRX_TX0	PASS	0.59
LTE_B38_Connected_V2_FBRX_PRX_TX0	PASS	0.91
LTE_B39_Connected_V2_FBRX_PRX_TX0	PASS	0.91
LTE_B40_Connected_V2_FBRX_PRX_TX0	PASS	0.59
LTE_B41_Connected_V2_FBRx_PRX_TX0	PASS	0.90

PRx	Sub6_n28_Connected_V2_Primary
	Sub6_n38_Connected_V2_Primary
	Sub6_n77A_Connected_V2_Primary_FBRx_PRx
	Sub6_n77B_Connected_V2_Primary_FBRx_PRx
DRx	Sub6_n1_Connected_V2_Diversity
	Sub6_n2_Connected_V2_Diversity
	Sub6_n3_Connected_V2_Diversity
Tx	Sub6_n1_UnConnected_V2
	LTE_B2_V2

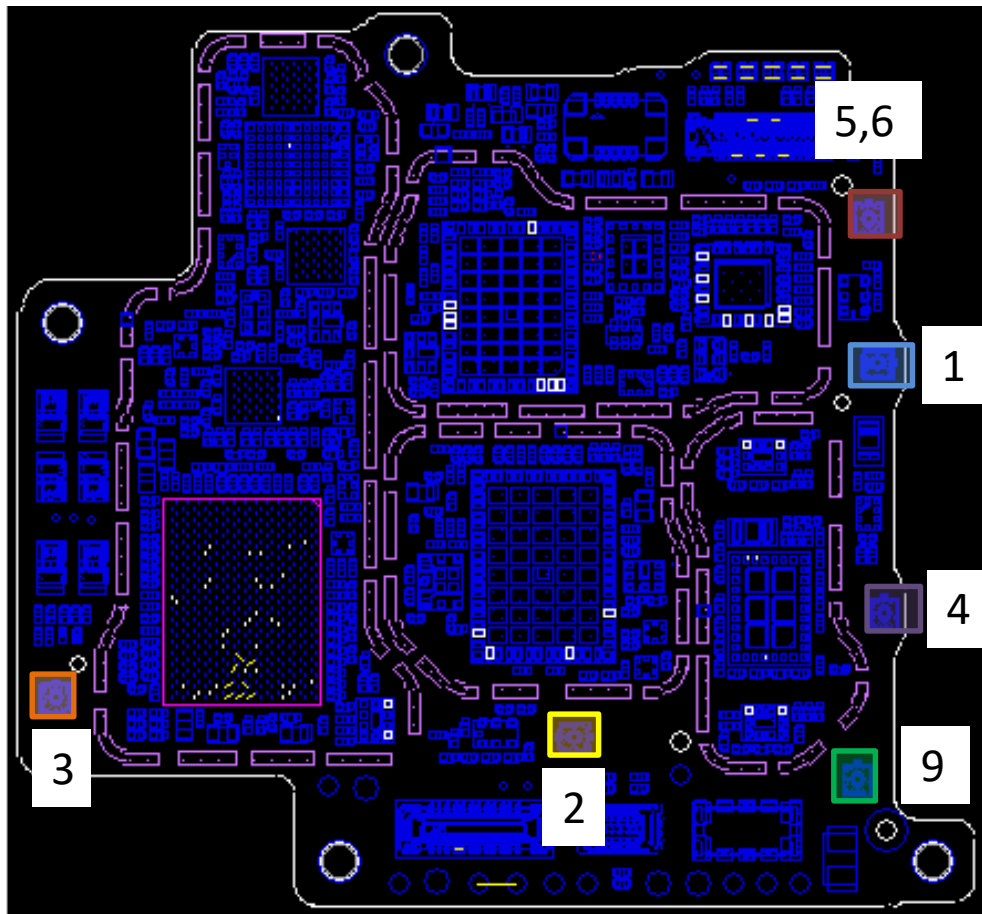
Search Cal log from SN computer. Find the corresponding hardware path according to the fail band. There is Cal log indication on left side ◦

Antenna Port

PARAMETERS: Rx Gain Calibration Parameters

Technology	Band	Sub Band	Signal Path	RFM Device	Antenna Path	PLL ID	Antenna Number	Time (s)
LTE	1	0	46	16	0	4	2	

bottom

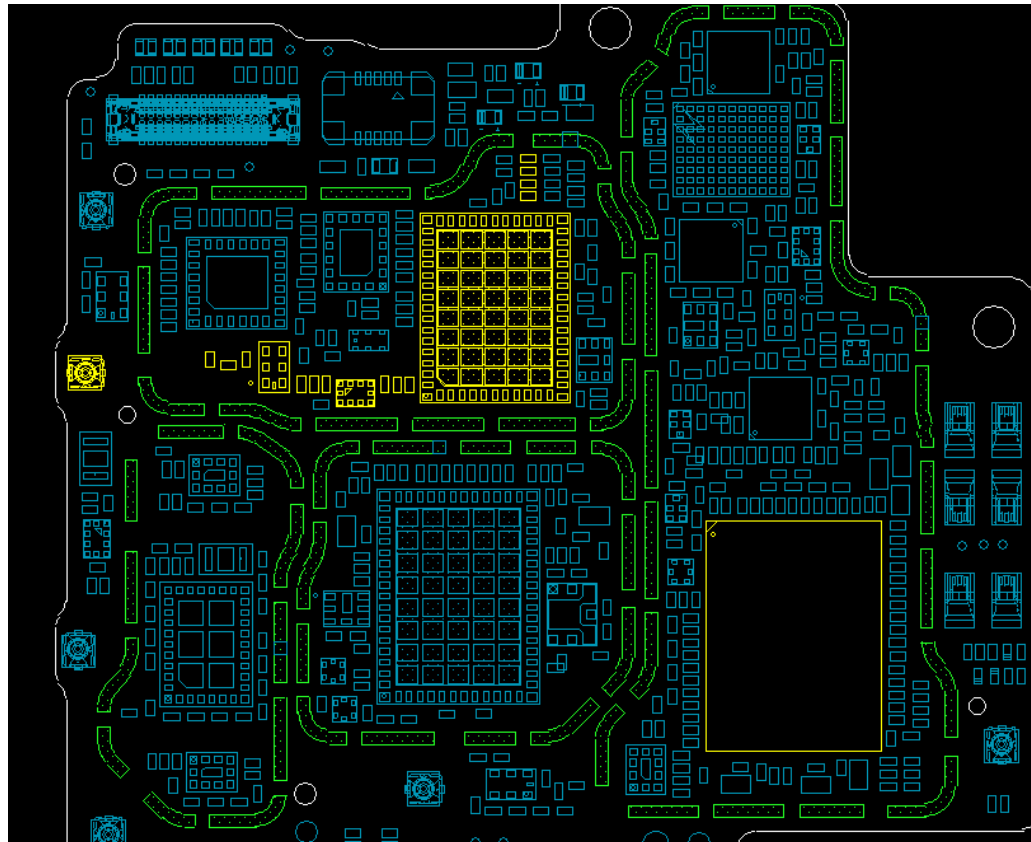


1. LB TRx & B1/B3/B7 PRx MIMO + NR
2. MHB TRx & N77 PRx MIMO + NR
3. LB DRx & B1/B3/B7 DRx MIMO + NR
4. N77 TRx
5. N77 DRx
6. MHB DRx & + NR
9. N77 DRx MIMO

GSM 850/900 Tx Path

bottom

RF Connector LB
CON4501



SDR888
U4101

LB + B1/B3/B7 DiPX
U4503

LB DPDT
U4502

LB PAMID QM77033D
U4501

GSM DCS/PCS Tx Path

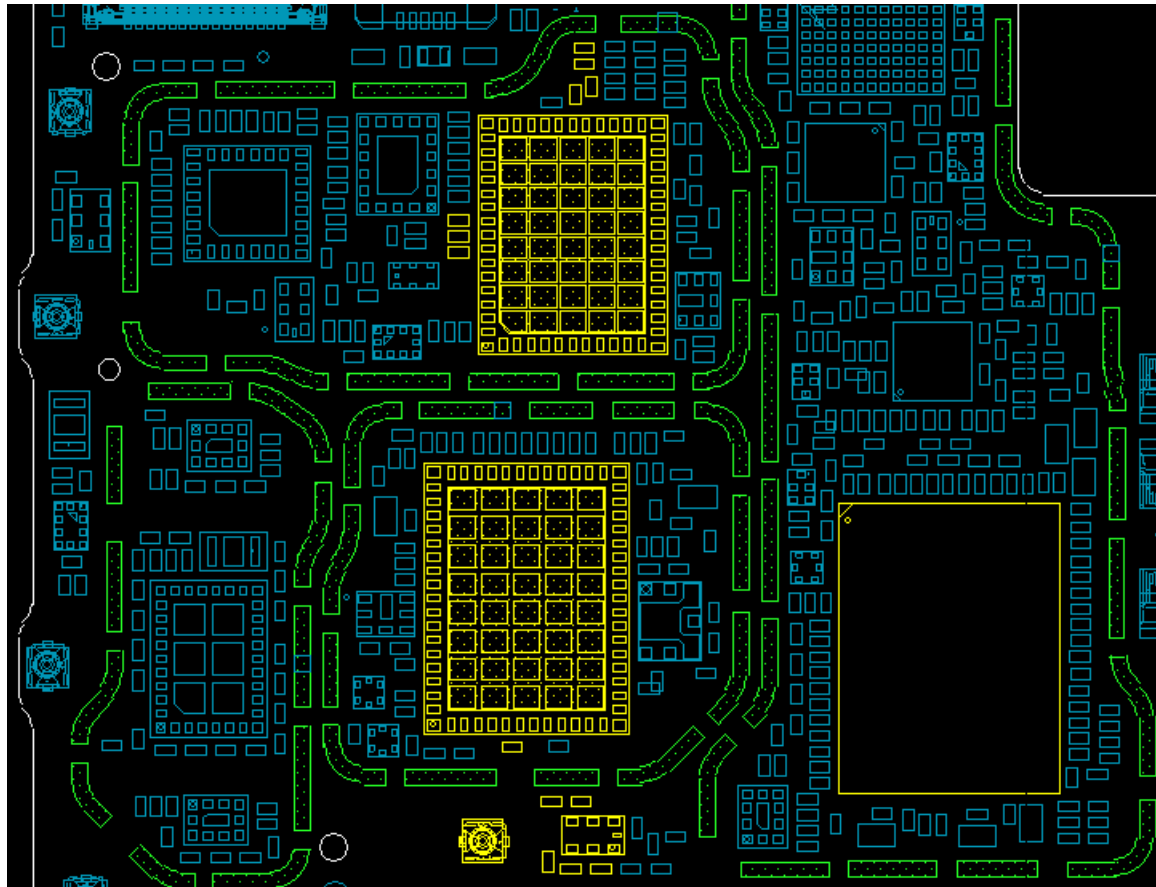
bottom

LB PAMID QM77033D

U4501

MHB PAMID QM77040

U4601



RF Connector MHB

CON4602

MHB + N77 DiPX

U4610

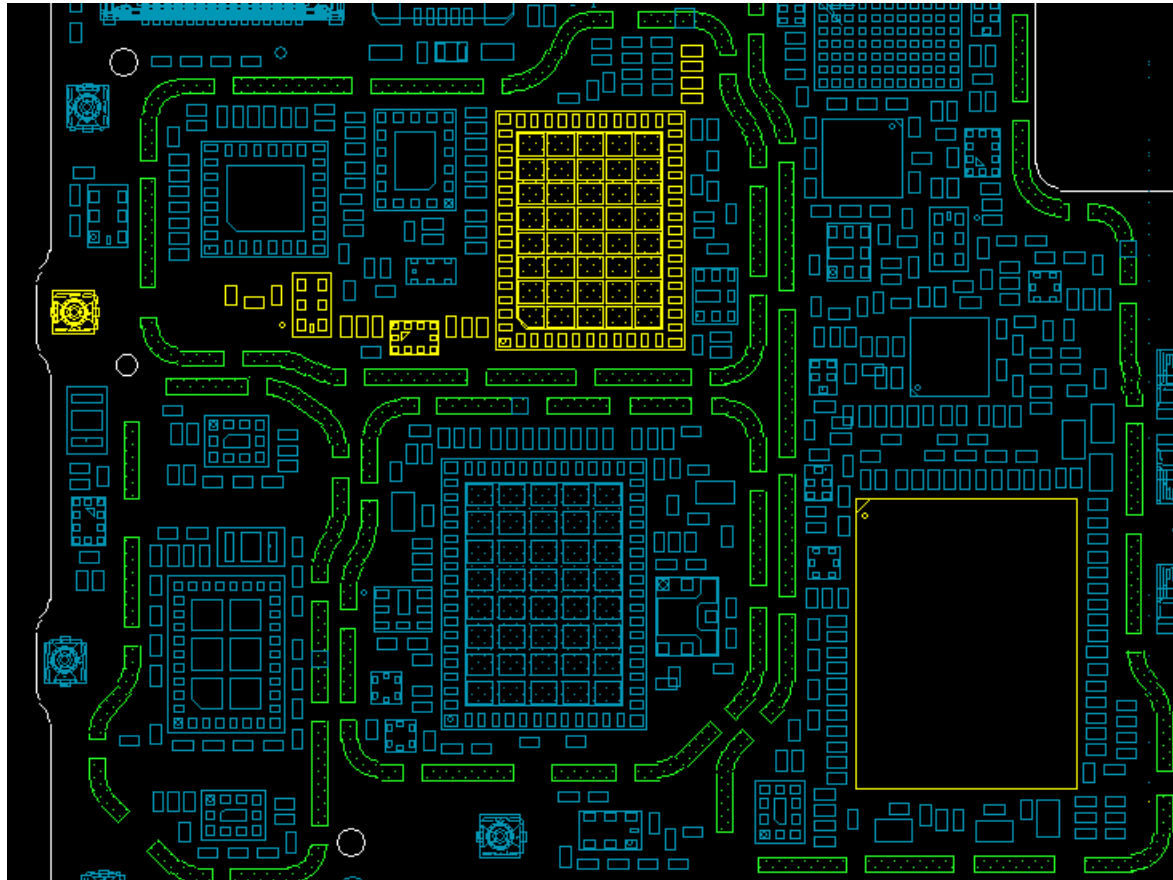
SDR888

U4101

3G/4G/5G LB Tx0 Path

bottom

RF Connector LB
CON4501



SDR888
U4101

LB + B1/B3/B7 DiPX
U4503

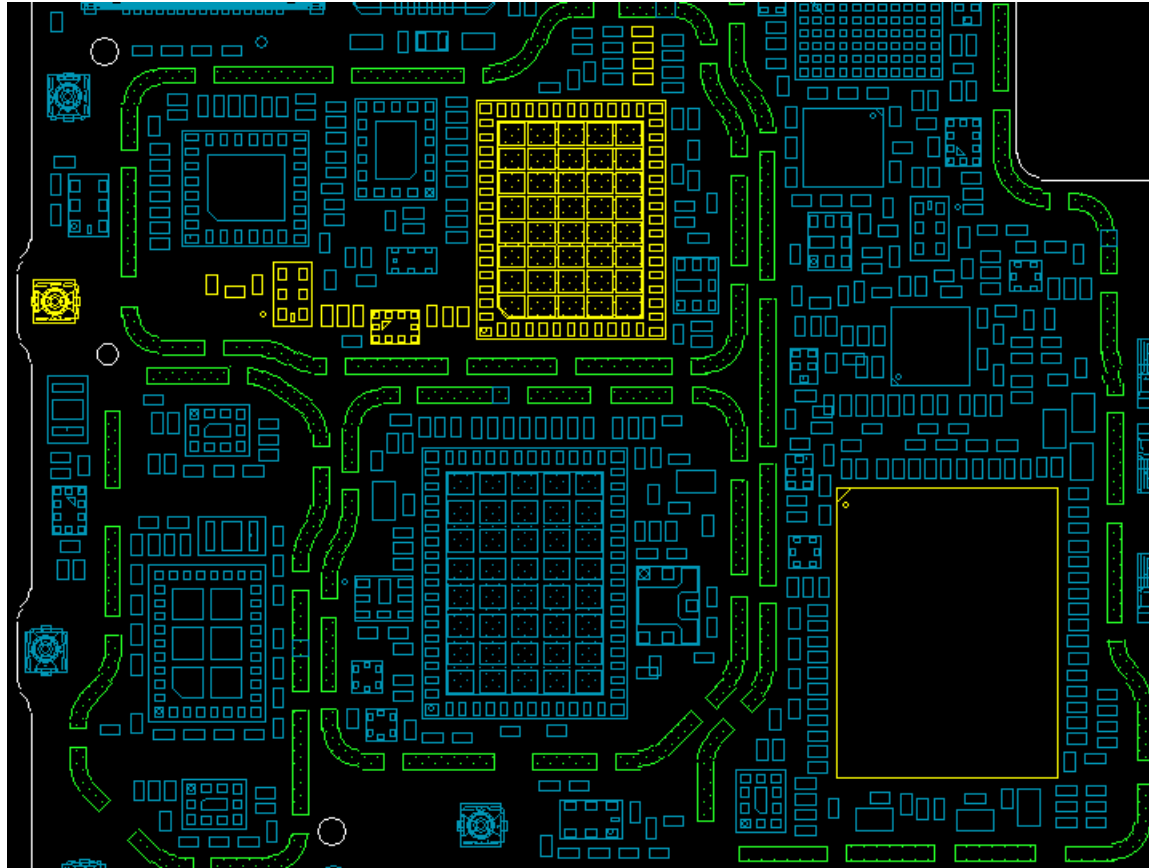
LB DPDT
U4502

LB PAMID QM77033D
U4501

3G/4G/5G LB Tx1 Path

bottom

RF Connector LB
CON4501



SDR888
U4101

LB + B1/B3/B7 DiPX
U4503

LB DPDT
U4502

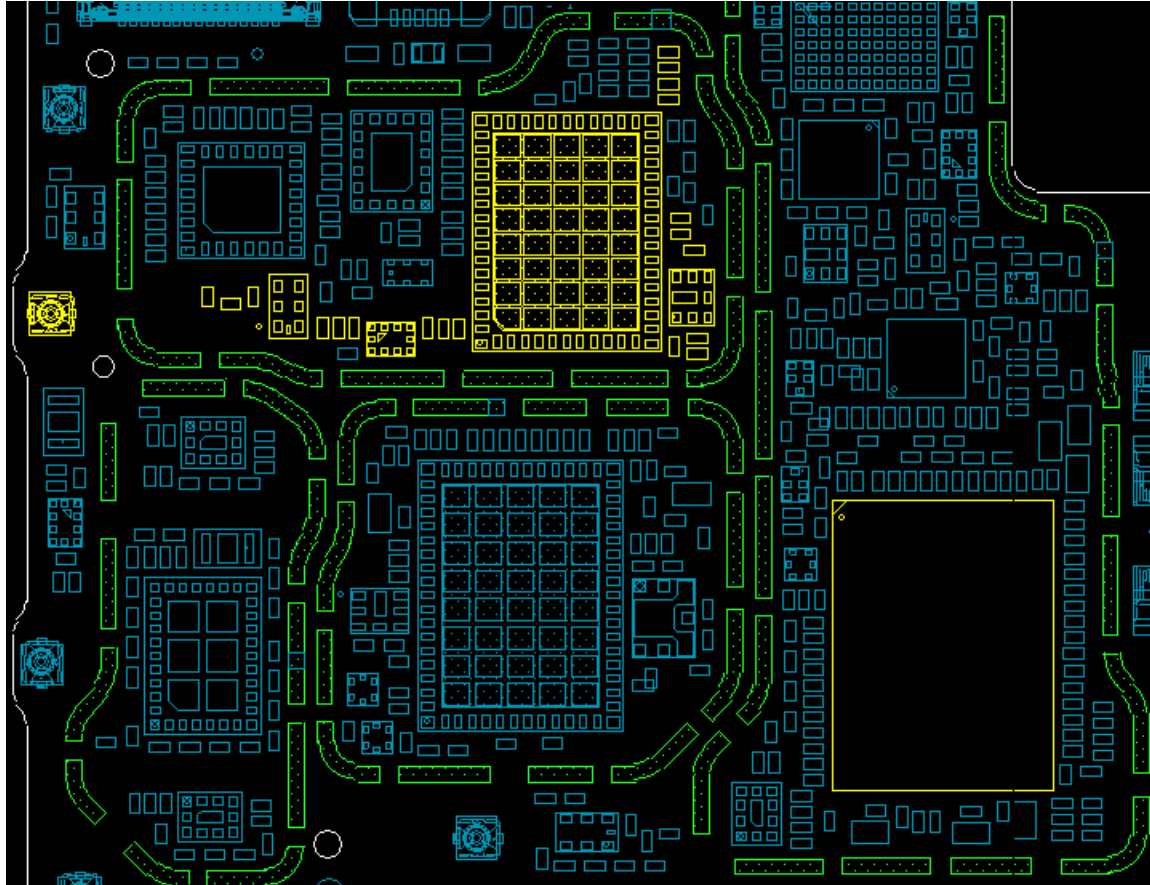
LB PAMID QM77033D
U4501

B28/B71 Tx0 Path

bottom

RF Connector LB
CON4501

LB + B1/B3/B7
DiPX
U4503



SDR888
U4101

LB DPDT
U4502

LB PAMID QM77033D
U4501

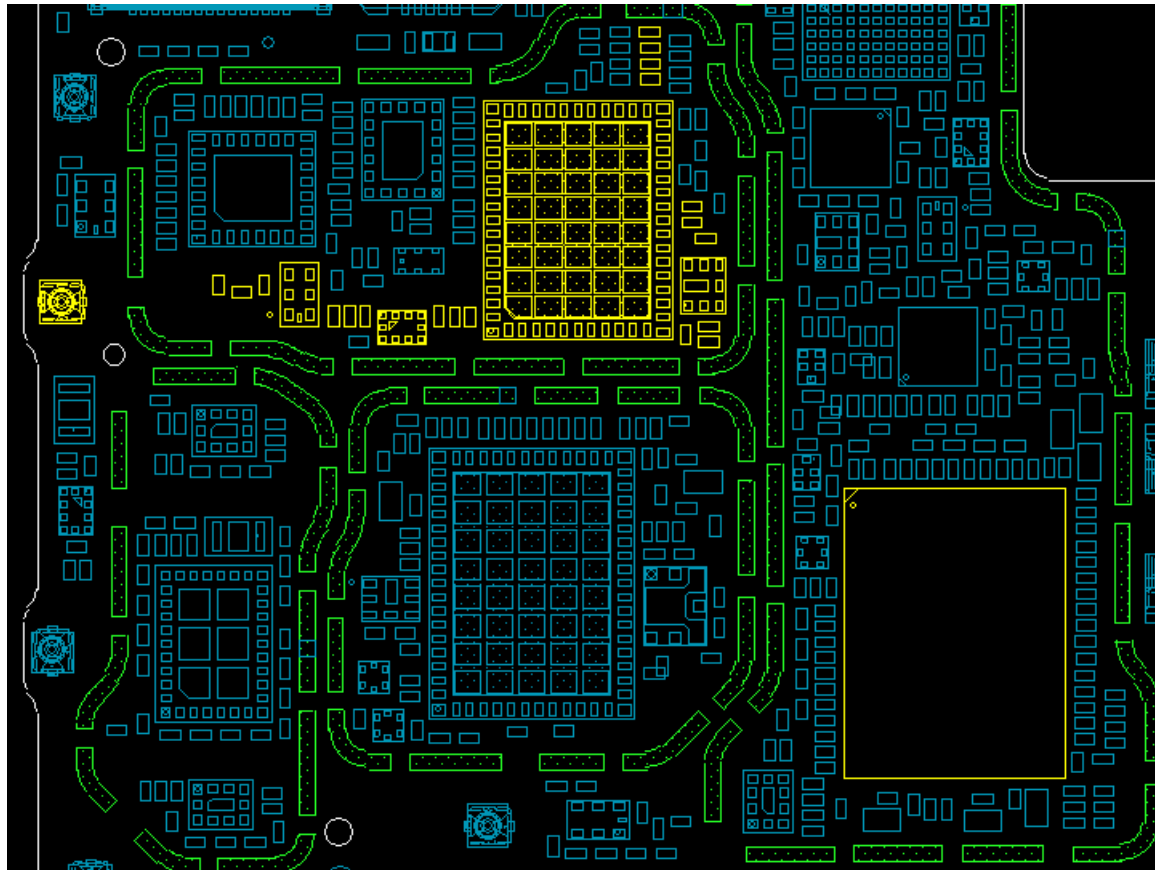
B28 DUX FILTER
U4504

B28/B71 Tx1 Path

bottom

RF Connector LB
CON4501

LB + B1/B3/B7
DiPX
U4503



SDR888
U4101

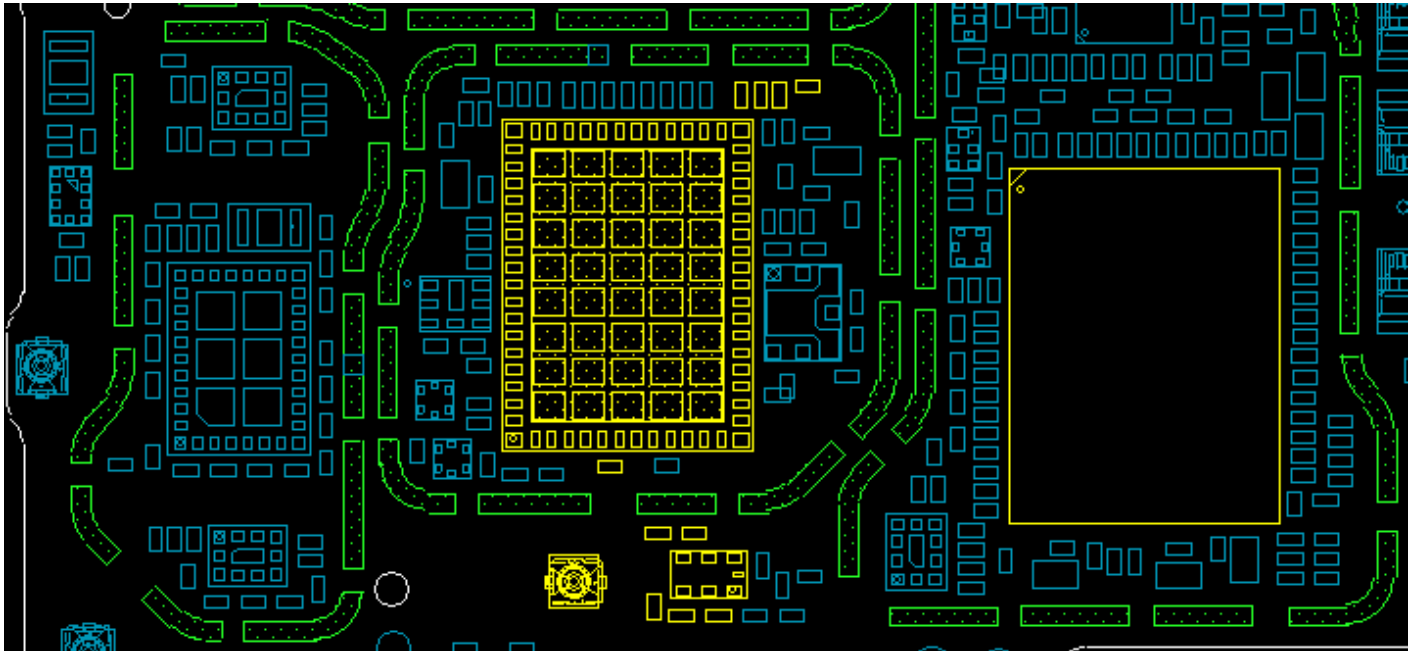
LB DPDT LB PAMID QM77033D
U4502 U4501

3G/4G/5G MB Tx Path

bottom

MHB PAMID QM77040

U4601



SDR888

U4101

RF Connector MHB

CON4602

MHB + N77 DiPX

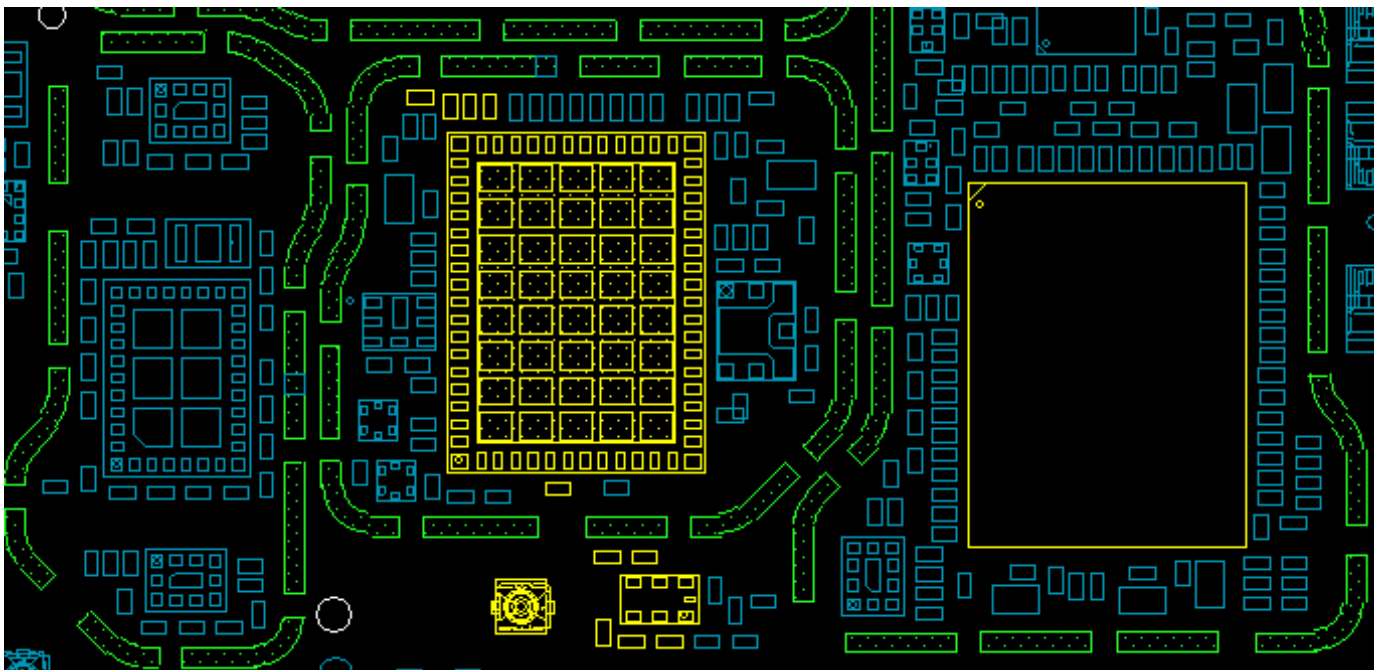
U4610

3G/4G/5G HB Tx Path

bottom

MHB PAMID QM77040

U4601



SDR888

U4101

RF Connector MHB

CON4602

MHB + N77 DiPX

U4610

B42/N77/N78 Tx Path

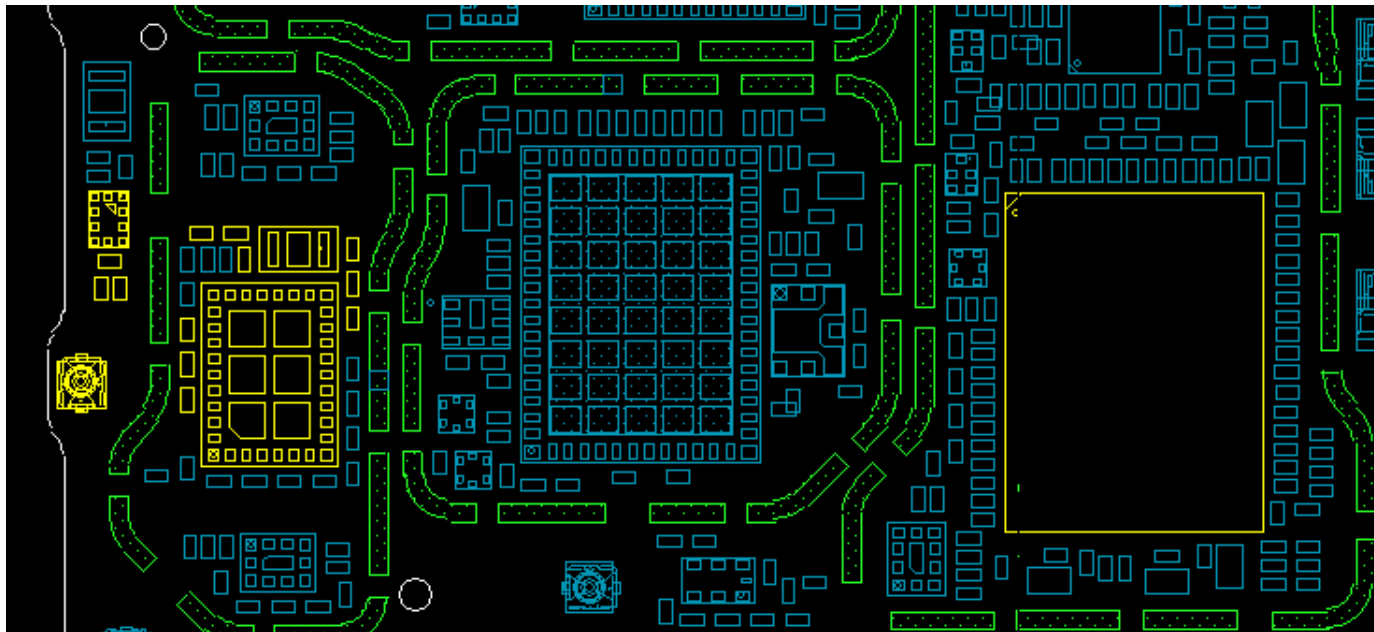
bottom

N77 BPF

U4705

N77 DPDT

U4702



SDR888

U4101

RF Connector N77

CON4701

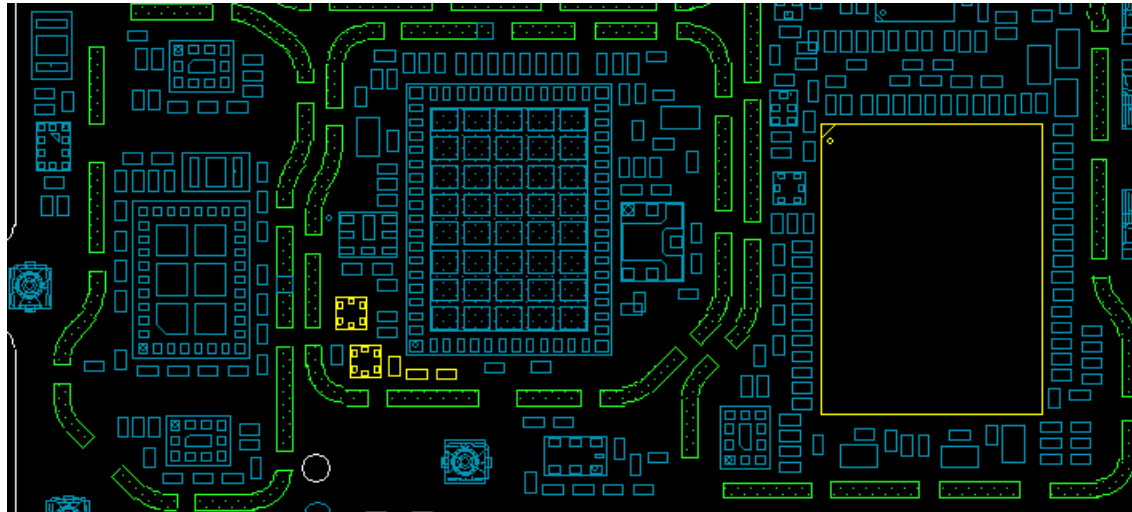
N77 PA QPM5677

U4701

FBRx Path

bottom

SPDT
U4611
SPDT
U4606



SDR888
U4101

DCDC

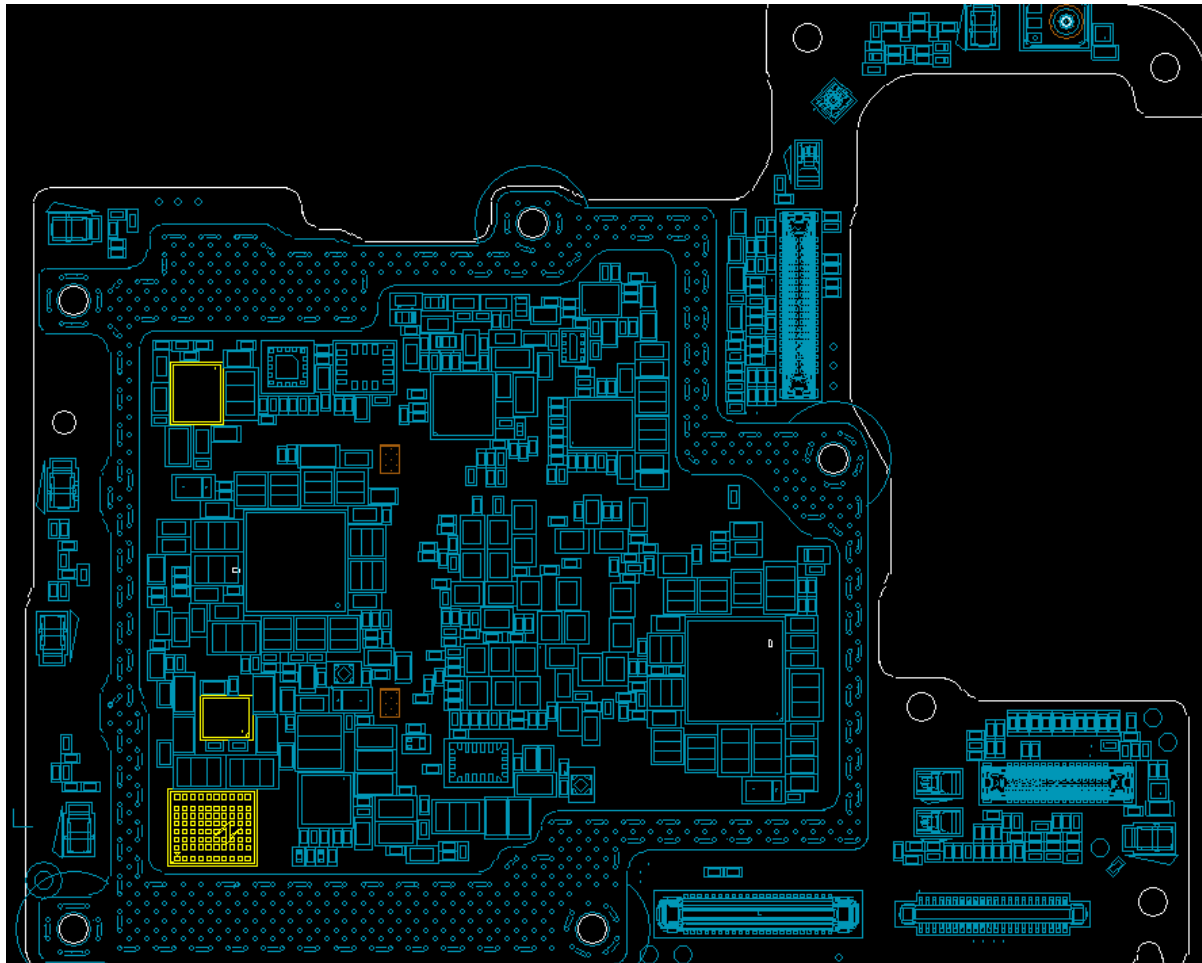
Bottom (MB)

DCDC位置在MB內側(bottom)，若要維修必須拆開interposer，無法拆解可直接換單板

QET5100
U4802

QET6100
U4901

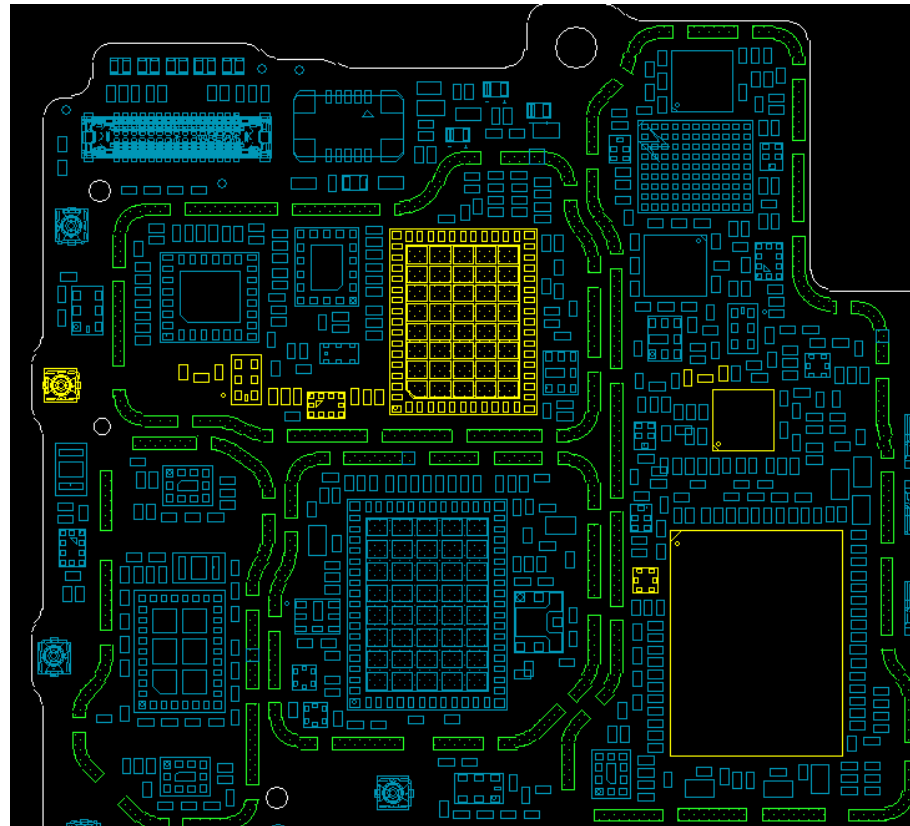
QET5100M
U4801



LB B8 PRx Path

bottom

RF Connector LB
CON4501



PRx LNA QLN5030
U5003

SDR888
U4101

LB + B1B3B7 DiPX
U4503

LB DPDT
U4502

LB PAMID QM77033D
U4501

LB LNA SPDT
U5007

LB B20 PRx Path

bottom

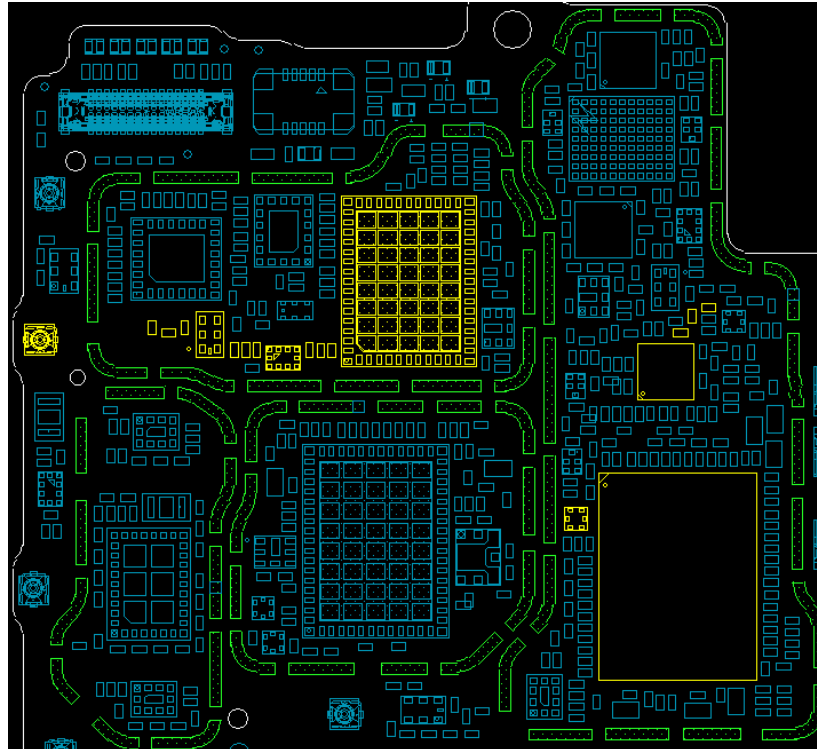
RF Connector LB
CON4501

LB + B1B3B7 DiPX
U4503

LB DPDT
U4502

LB PAMID QM77033D
U4501

LB LNA SPDT
U5007

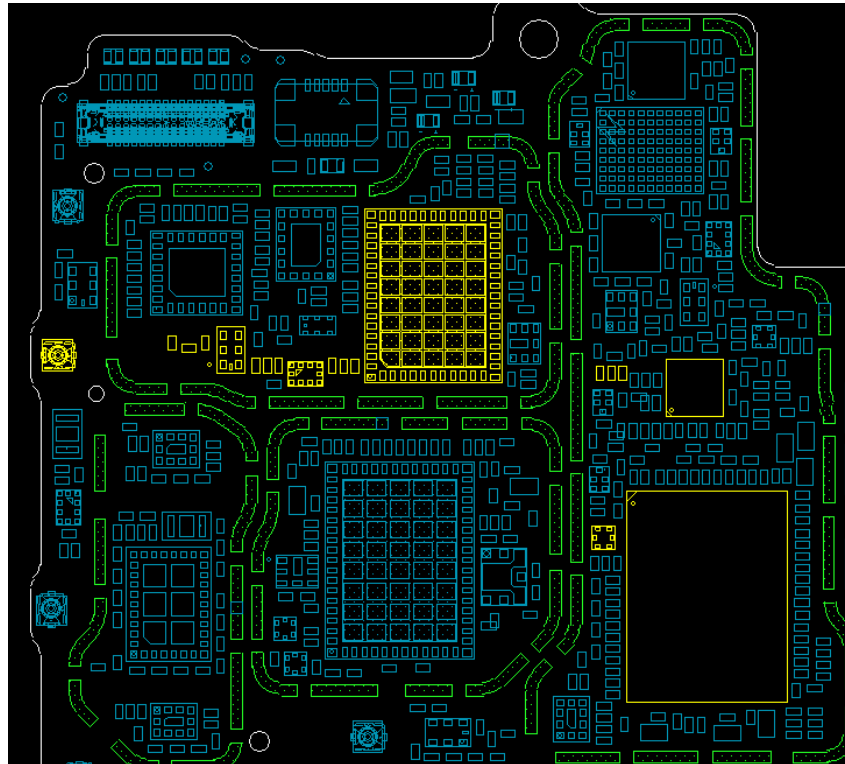


PRx LNA QLN5030
U5003

SDR888
U4101

LB B26 PRx Path

bottom



RF Connector LB
CON4501

PRx LNA QLN5030
U5003

SDR888
U4101

LB + B1B3B7 DiPX
U4503

LB PAMID QM77033D
U4501

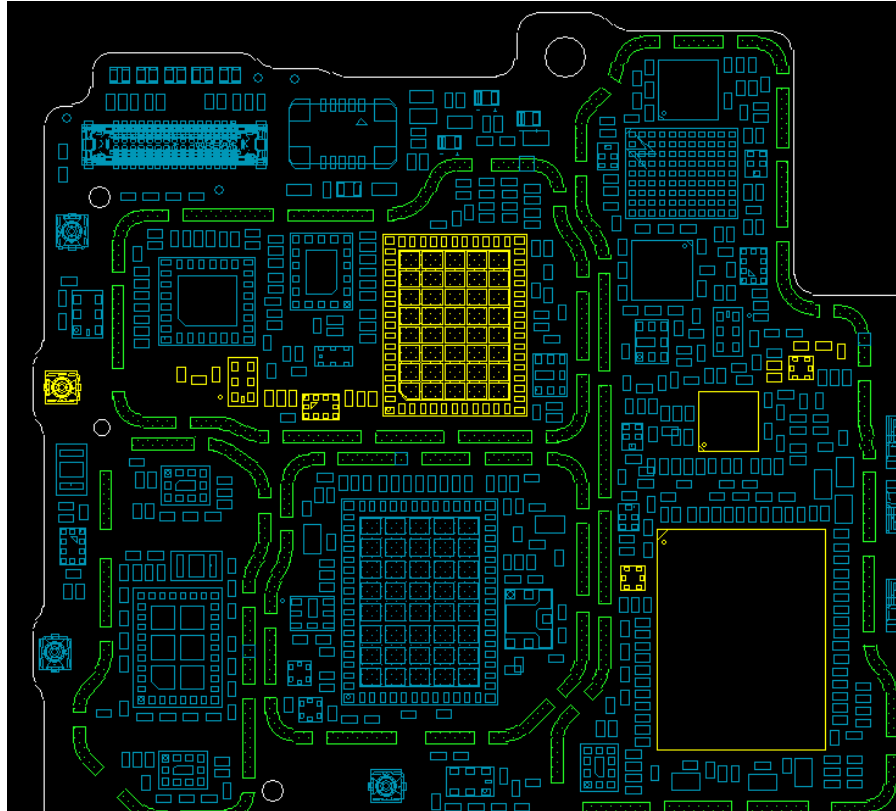
LB DPDT
U4502

LB LNA SPDT
U5007

LB B12 PRx Path

bottom

RF Connector LB
CON4501



SPDT

U5001

PRx LNA QLN5030

U5003

SDR888

U4101

LB + B1B3B7 DiPX

U4503

LB PAMID QM77033D

U4501

LB DPDT

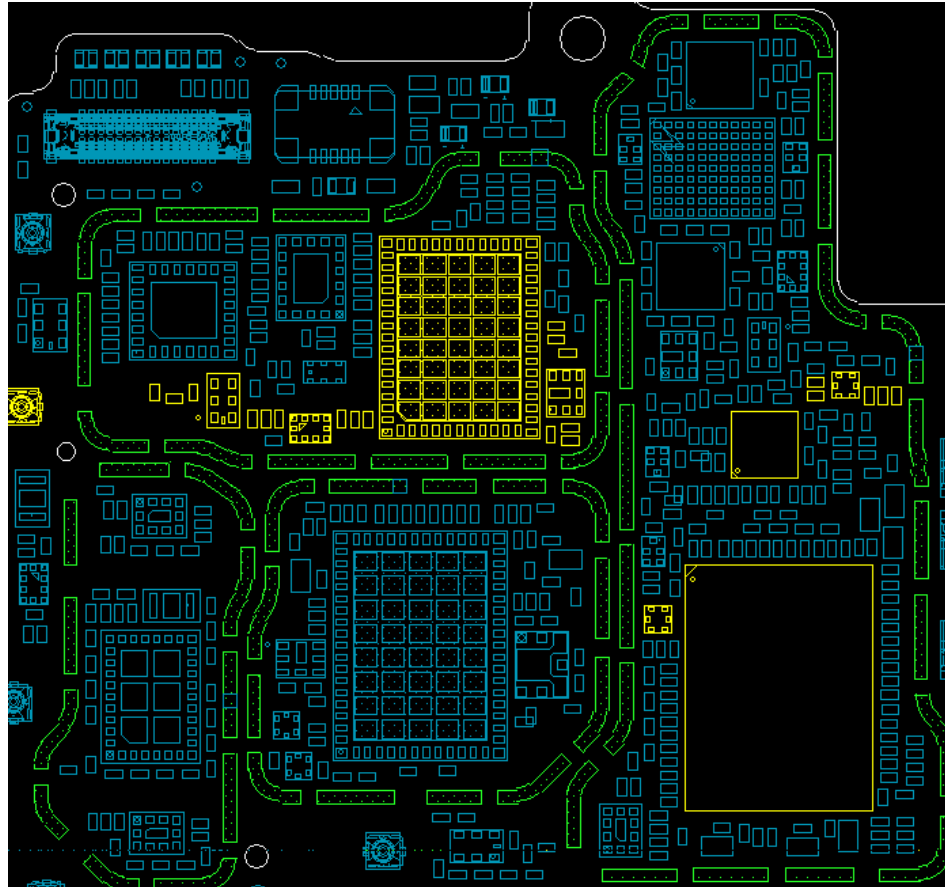
U4502

LB LNA SPDT

U5007

LB B28 PRx Path (WW)

bottom



RF Connector LB
CON4501

SPDT
U5001

PRx LNA QLN5030
U5003

SDR888
U4101

LB + B1B3B7 DiPX
U4503

LB PAMID QM77033D
U4501

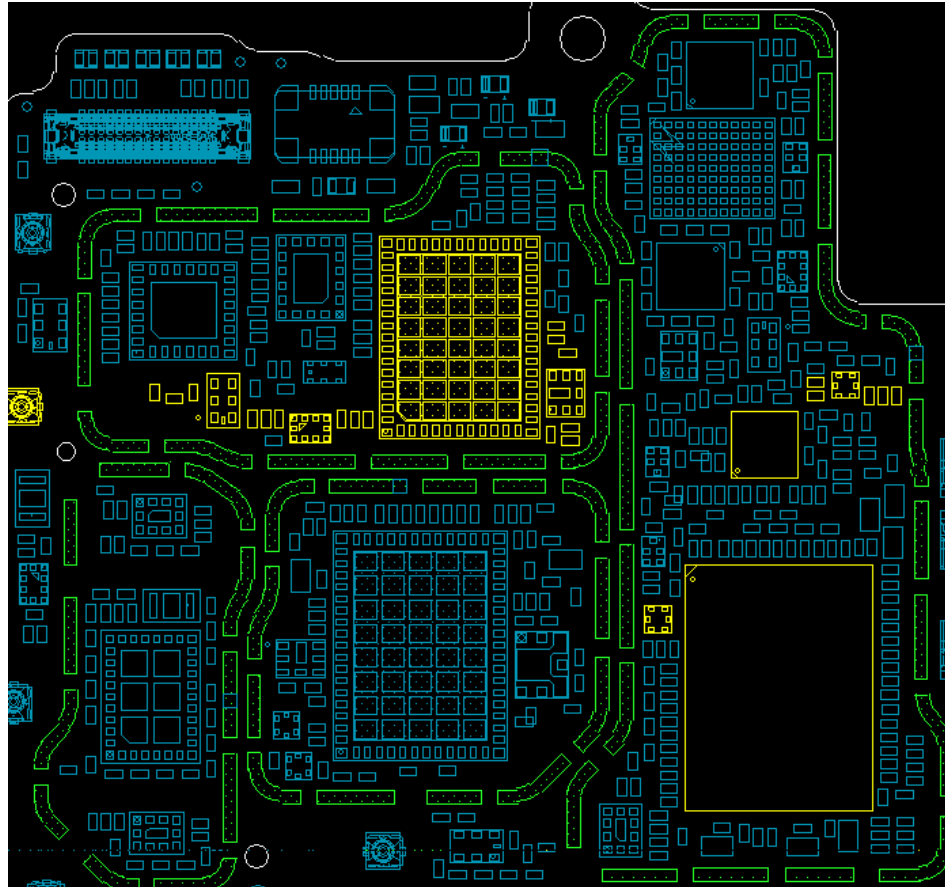
LB DPDT
U4502

B28 DUP
U4504

LB LNA SPDT
U5007

LB B71 PRx Path (US)

bottom



RF Connector LB
CON4501

SPDT
U5001

PRx LNA QLN5030
U5003

SDR888
U4101

LB + B1B3B7 DiPX
U4503

LB DPDT
U4502

LB PAMID QM77033D
U4501

B71 DUP
U4504

LB LNA SPDT
U5007

MB B1/B4 PRx Path

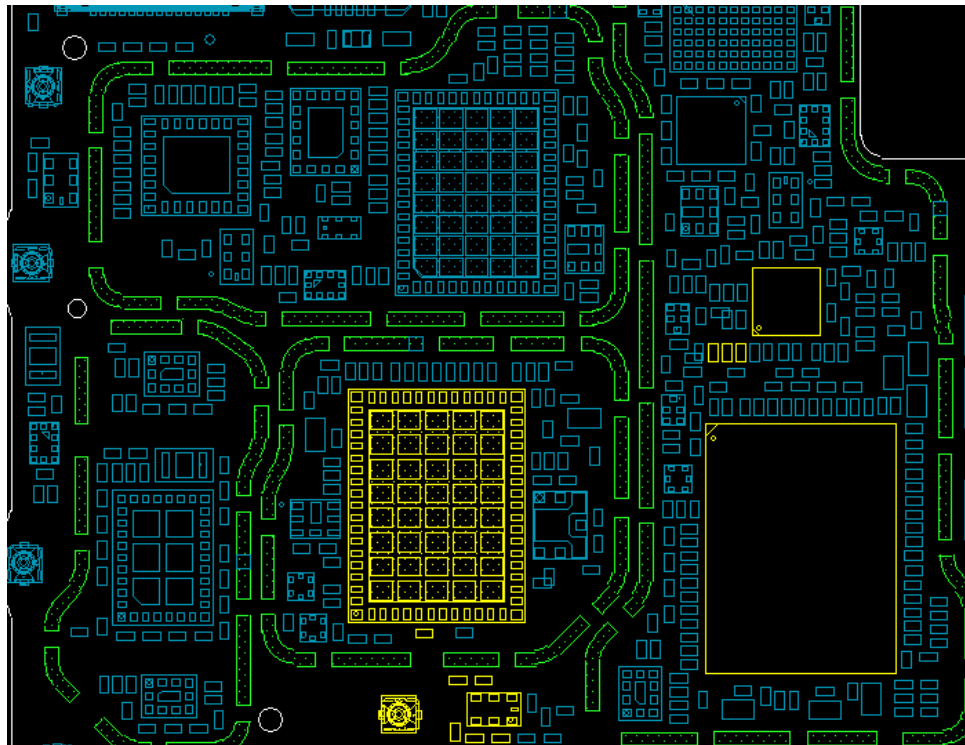
bottom

WW: QM77040

US: QM77038

MHB PAMID QM77040

U4601



PRx LNA QLN5030

U5003

SDR888

U4101

RF Connector MHB

CON4602

MHB + N77 DiPX

U4610

MB B2 PRx Path(WW)

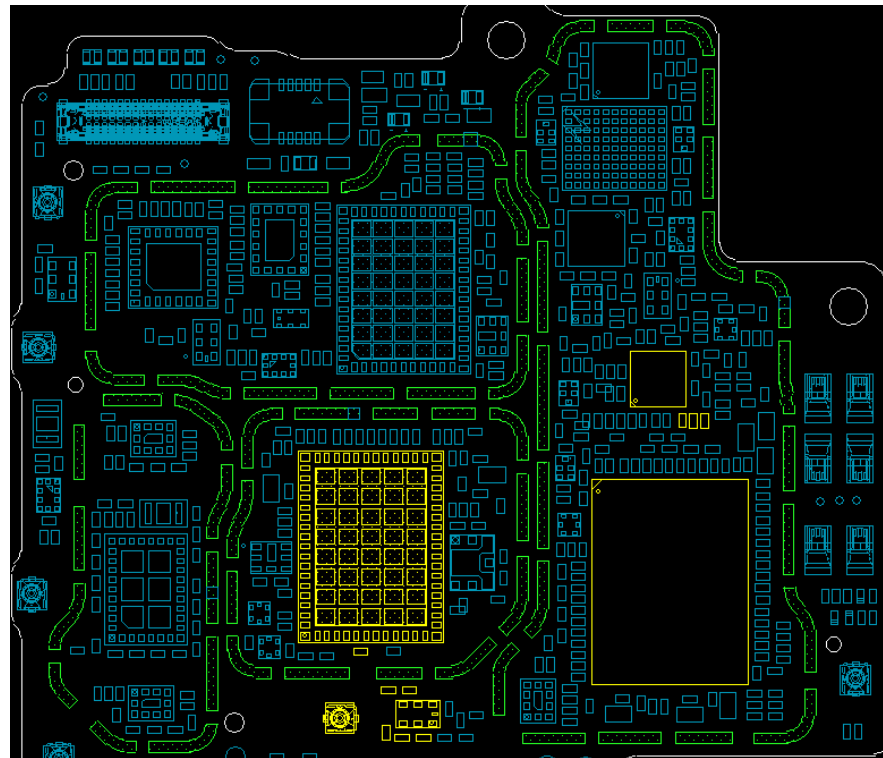
bottom

WW: QM77040

US: QM77038

MHB PAMID QM77040

U4601



PRx LNA QLN5030

U5003

SDR888

U4101

RF Connector MHB

CON4602

MHB + N77 DiPX

U4610

MB B2/B25 PRx Path (US)

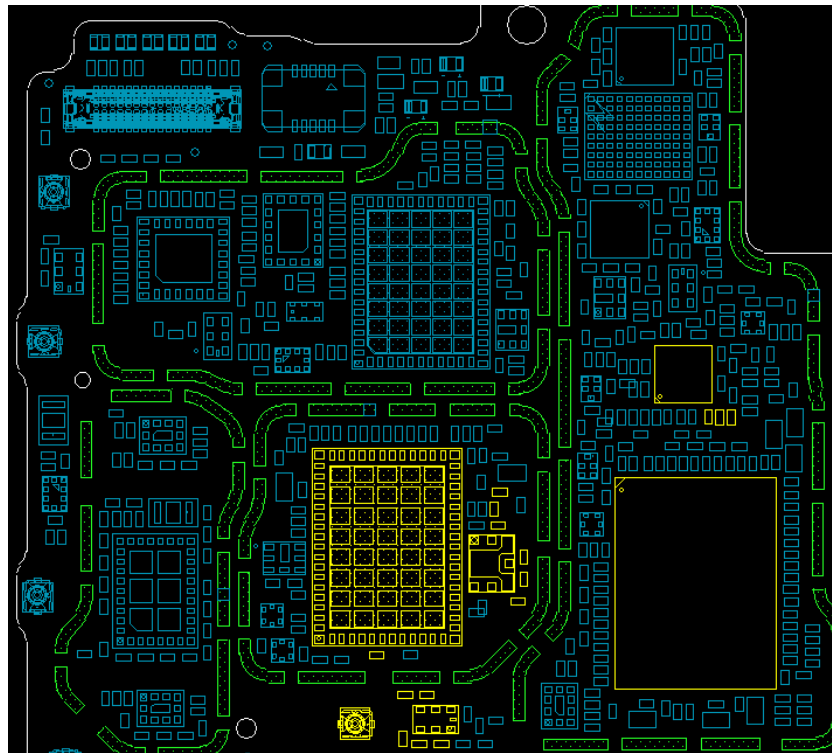
bottom

WW: QM77040

US: QM77038

MHB PAMID QM77038

U4601



PRx LNA QLN5030

U5003

SDR888

U4101

RF Connector MHB

CON4602

MHB + N77 DiPX

U4610

Q-PLX

U4607

MB B3 PRx Path

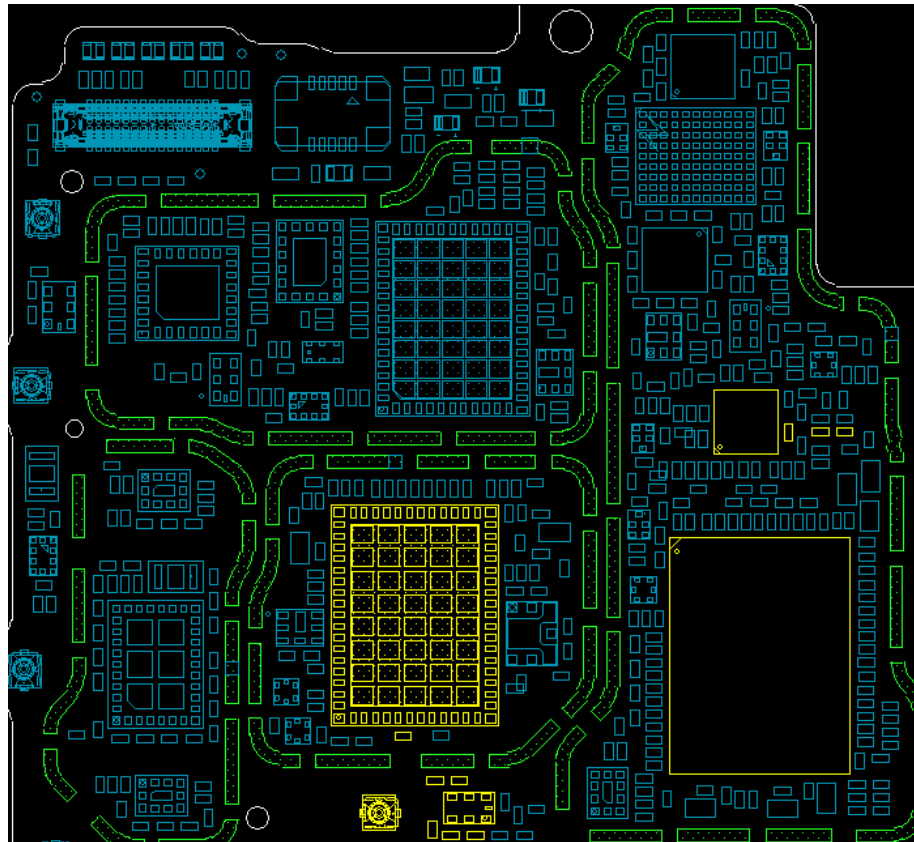
bottom

MHB PAMID QM77040

U4601

WW: QM77040

US: QM77038



PRx LNA QLN5030

U5003

SDR888

U4101

RF Connector MHB

CON4602

MHB + N77 DiPX

U4610

MB B39 PRx Path

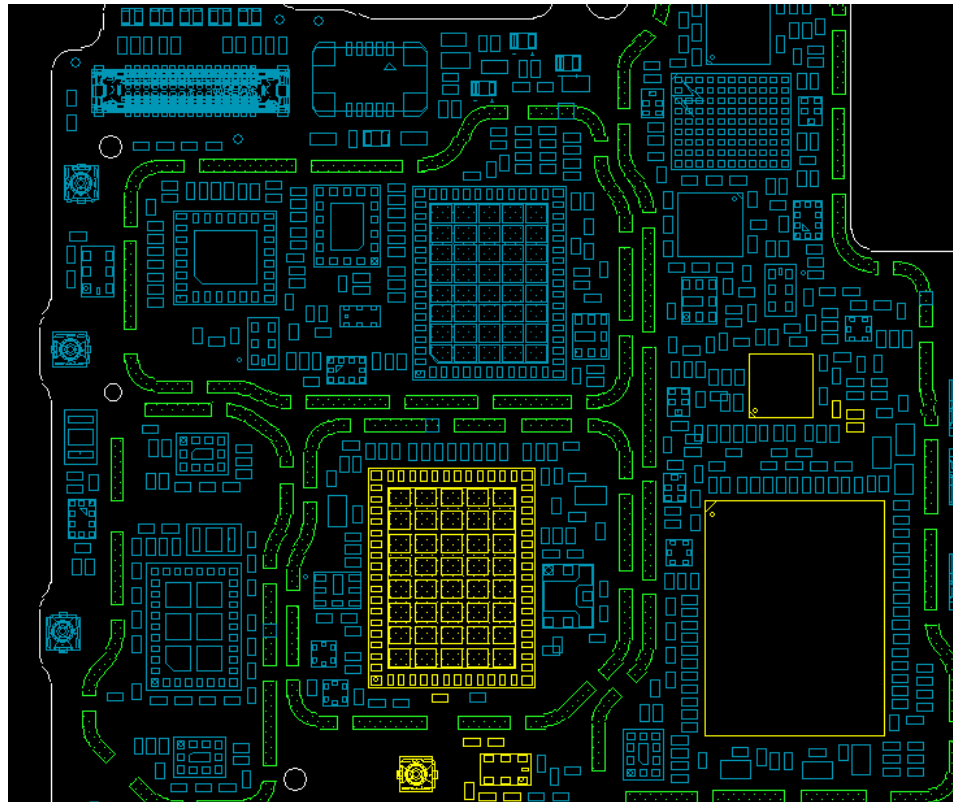
bottom

MHB PAMID QM77040

U4601

WW: QM77040

US: QM77038



PRx LNA QLN5030

U5003

SDR888

U4101

RF Connector MHB

CON4602

MHB + N77 DiPX

U4610

MB B34 PRx Path (WW)

bottom

WW: QM77040

US: QM77038

B34 FILTER

U5002

MHB PAMID QM77040

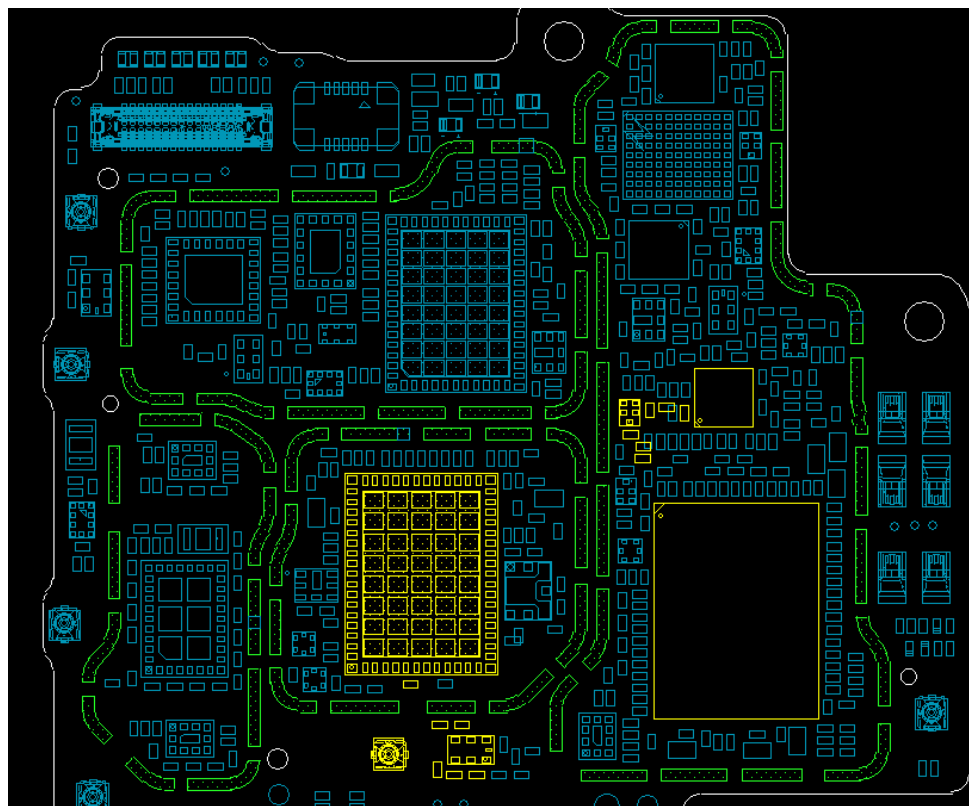
U4601

PRx LNA QLN5030

U5003

SDR888

U4101



RF Connector MHB

CON4602

MHB + N77 DiPX

U4610

MB B34 PRx Path (US)

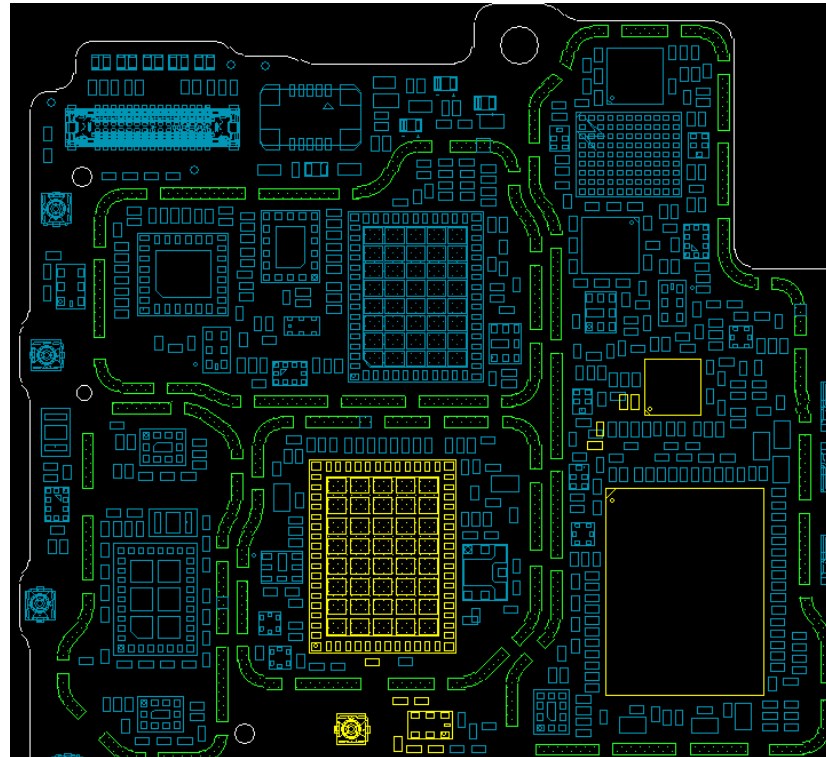
bottom

WW: QM77040

US: QM77038

MHB PAMID QM77038

U4601



PRx LNA QLN5030

U5003

SDR888

U4101

RF Connector MHB

CON4602

MHB + N77 DiPX

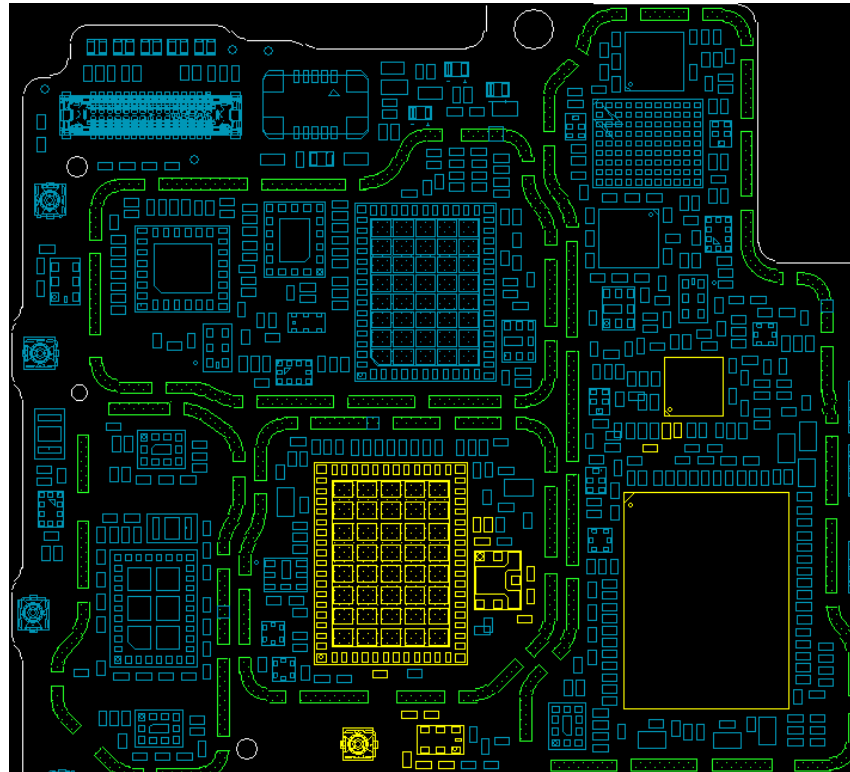
U4610

MB B66 PRx Path (US)

WW: QM77040

US: QM77038

bottom



MHB PAMID QM77038

U4601

PRx LNA QLN5030

U5003

SDR888

U4101

RF Connector MHB

CON4602

MHB + N77 DiPX

U4610

Q-PLX

U4607

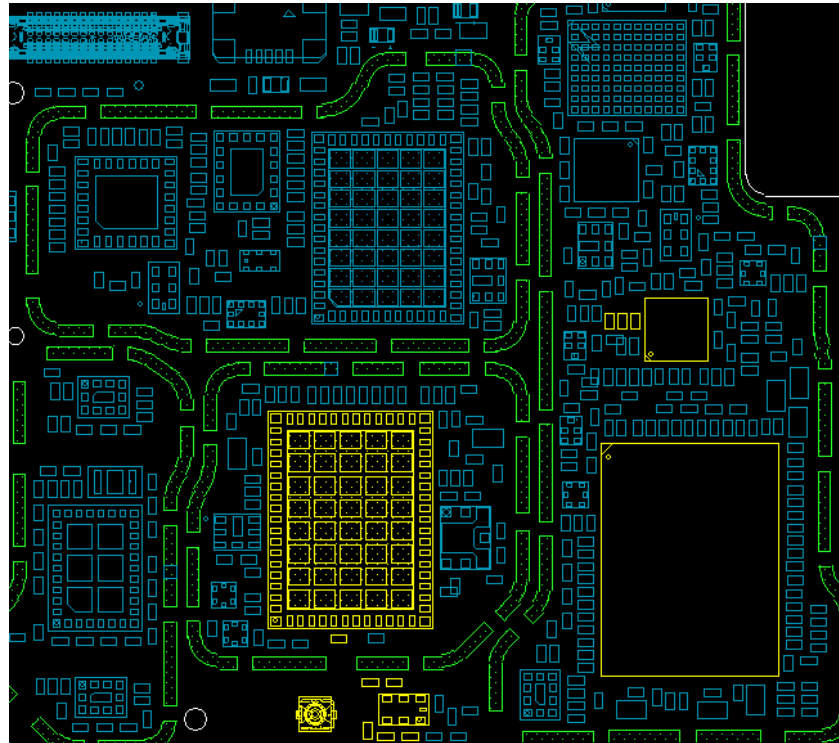
HB B7 PRx Path

bottom

WW: QM77040

US: QM77038

MHB PAMID QM77040
U4601



PRx LNA QLN5030
U5003

SDR888
U4101

RF Connector MHB MHB + N77 DiPX
CON4602 **U4610**

HB B40 PRx Path

bottom

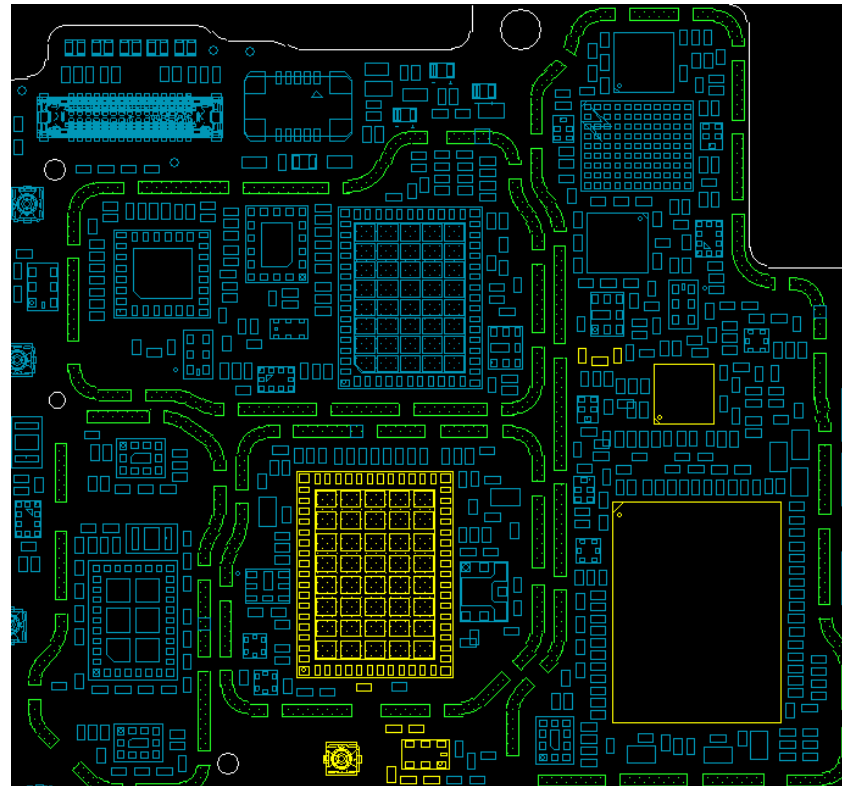
WW: QM77040

US: QM77038

MHB PAMID QM77040
U4601

PRx LNA QLN5030
U5003

SDR888
U4101



RF Connector MHB
CON4602

MHB + N77 DiPX
U4610

HB B38/B41 PRx Path

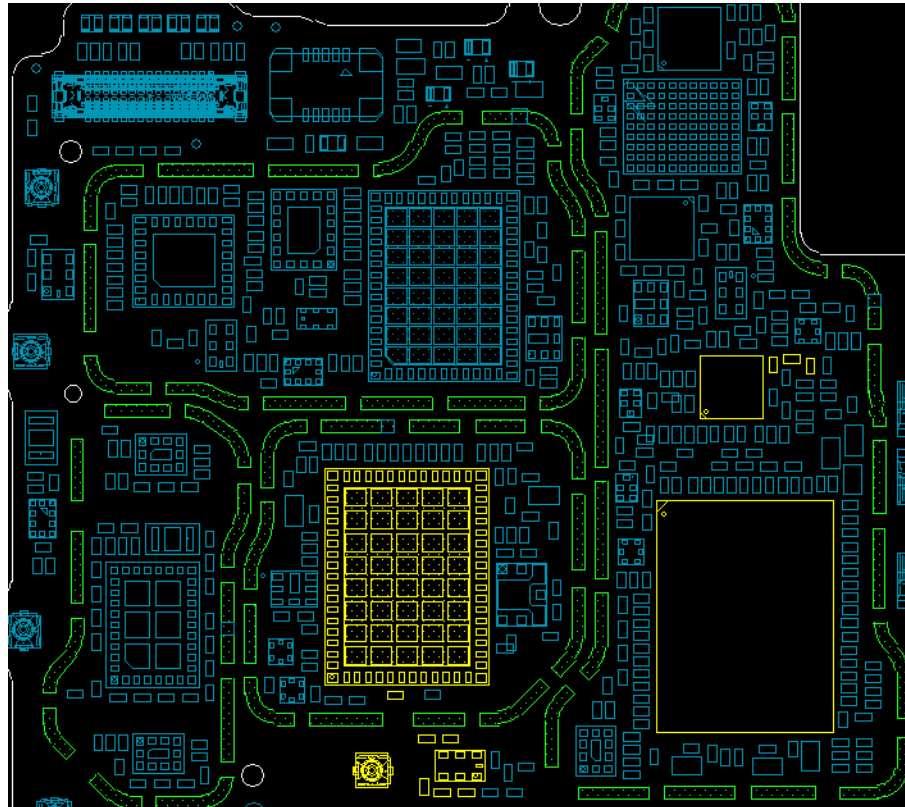
bottom

WW: QM77040

US: QM77038

MHB PAMID QM77040

U4601



PRx LNA QLN5030

U5003

SDR888

U4101

RF Connector MHB

CON4602

MHB + N77 DiPX

U4610

MB B30 PRx Path (US)

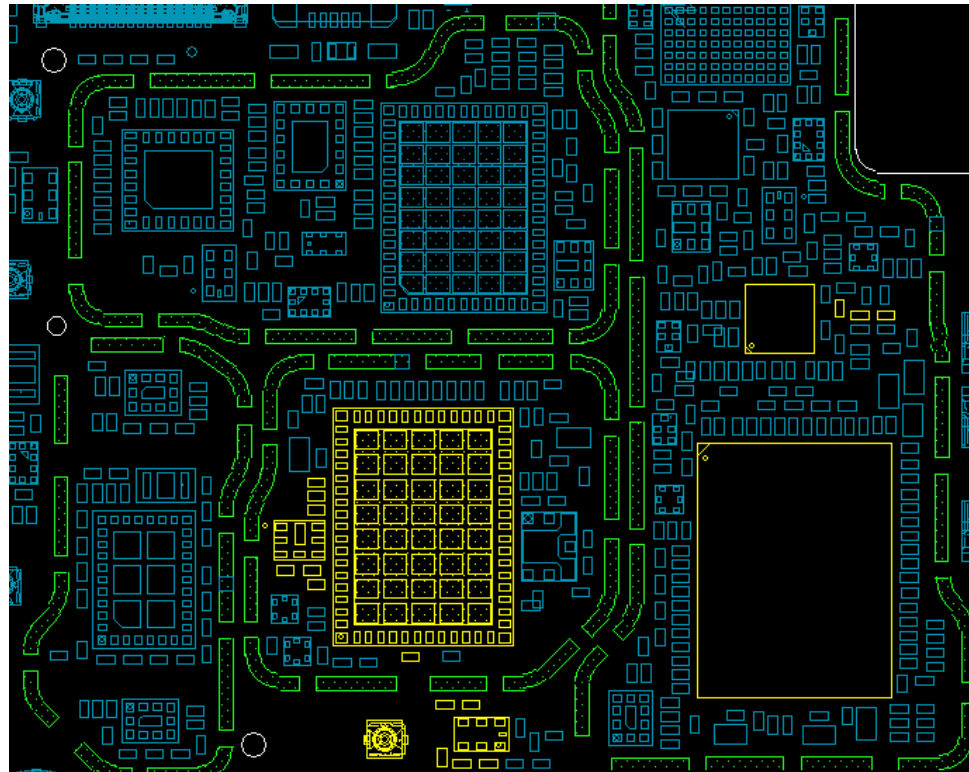
bottom

WW: QM77040

US: QM77038

MHB PAMID QM77038

U4601



DU-PLX

U4605

PRx LNA QLN5030

U5003

SDR888

U4101

RF Connector MHB

CON4602

MHB + N77 DiPX

U4610

B42/N77/N78 PRx Path

bottom

N77 BPF

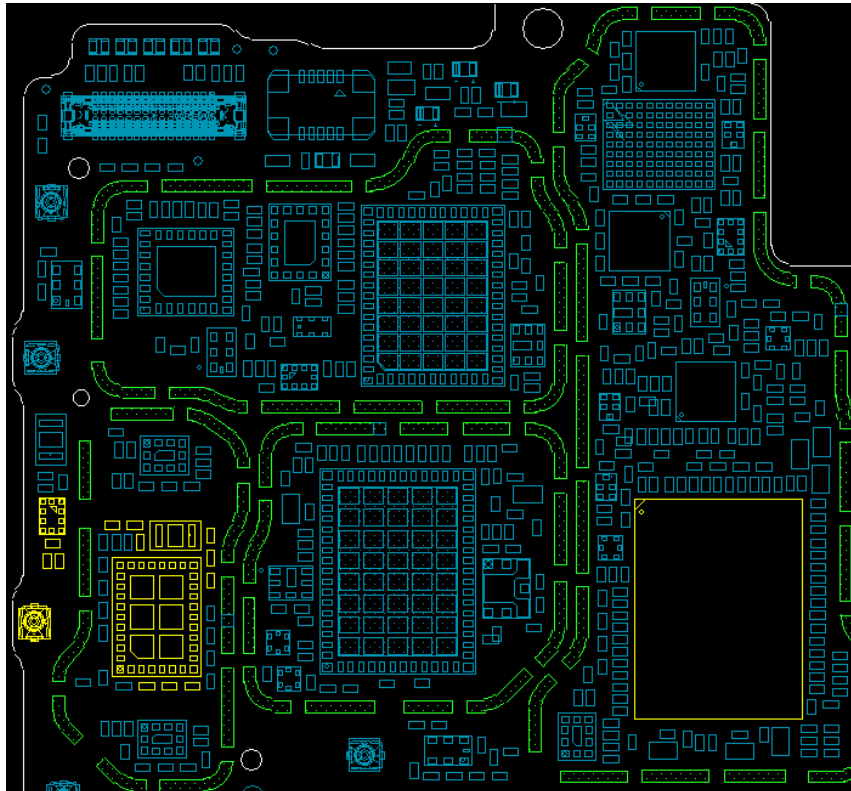
U4705

N77 DPDT

U4702

RF Connector N77

CON4701



SDR88

U4101

N77 PA QPM5677

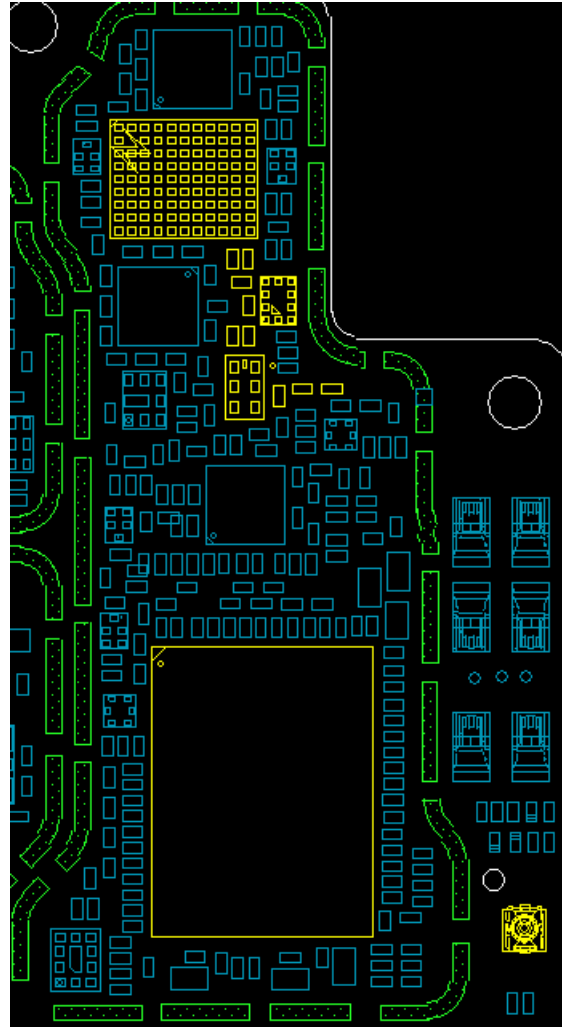
U4701

LB DRx Path

bottom

DiFEM QDM3302
U5206

SDR888
U4101



LB DRx DPDT
U5205

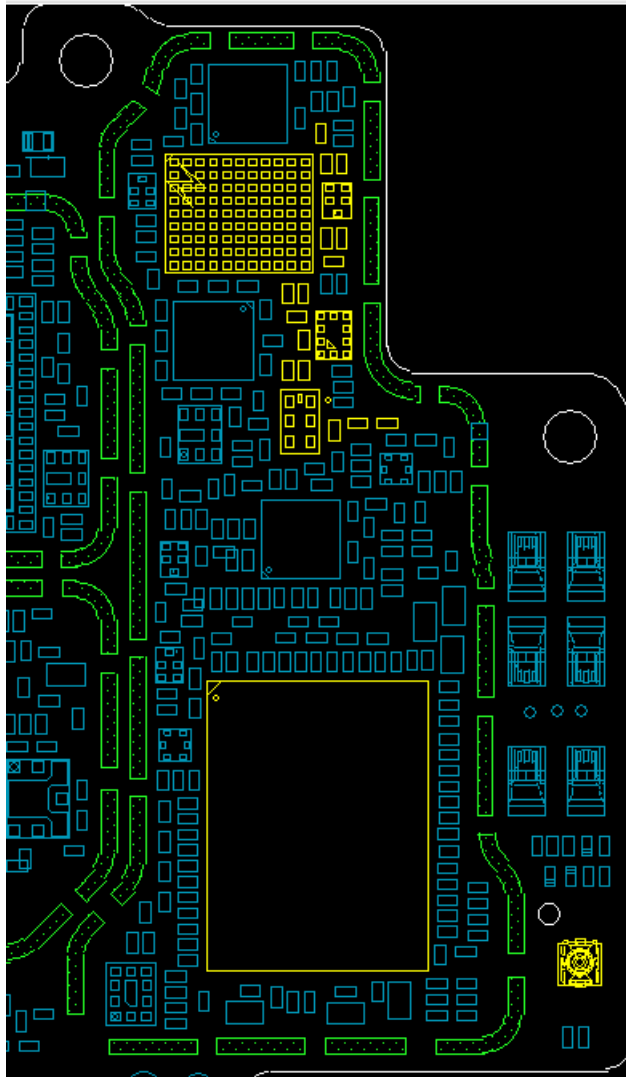
LB+B1B3B7 DiPX
U5202

Div. RF connector LB
CON5204

B28/B71 DRx Path

DiFEM QDM3302
U5206

SDR888
U4101



filter
U5204

LB DRx DPDT
U5205

LB+B1B3B7 DiPX
U5202

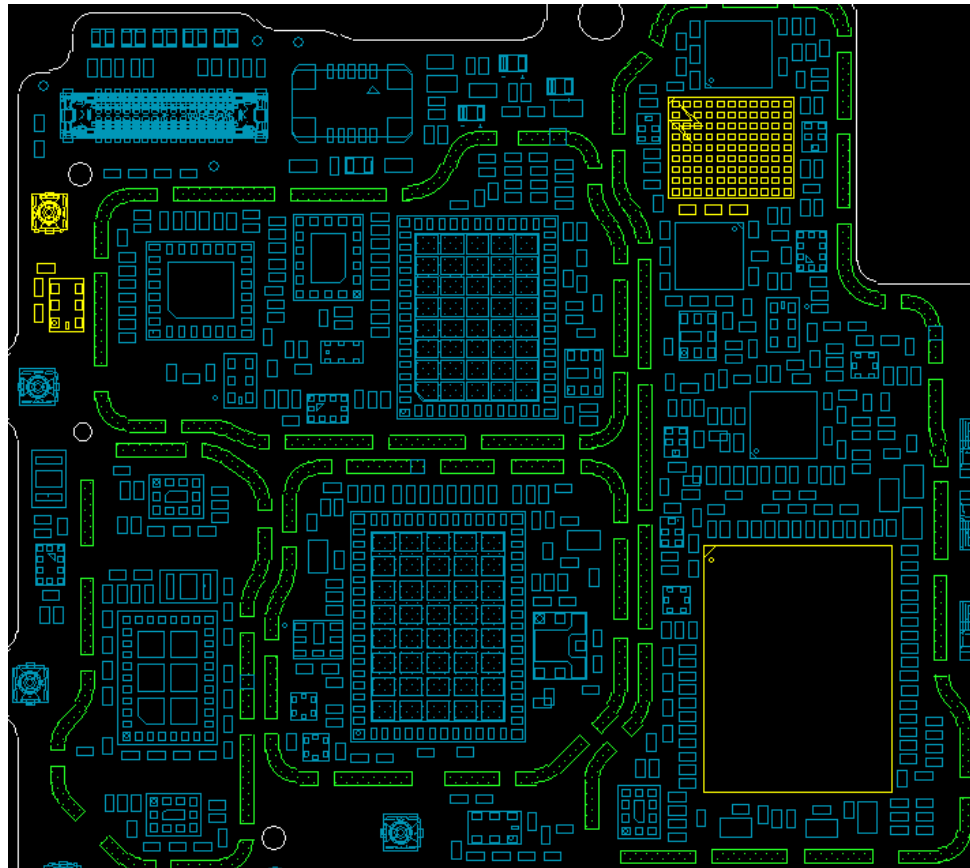
Div. RF connector LB
CON5204

MHB DRx Path

bottom

Div. RF
connector MHB
CON5205

MHB di-plx
U5203



DiFEM QDM3302
U5206

SDR888
U4101

B42/N77/N78 DRx Path

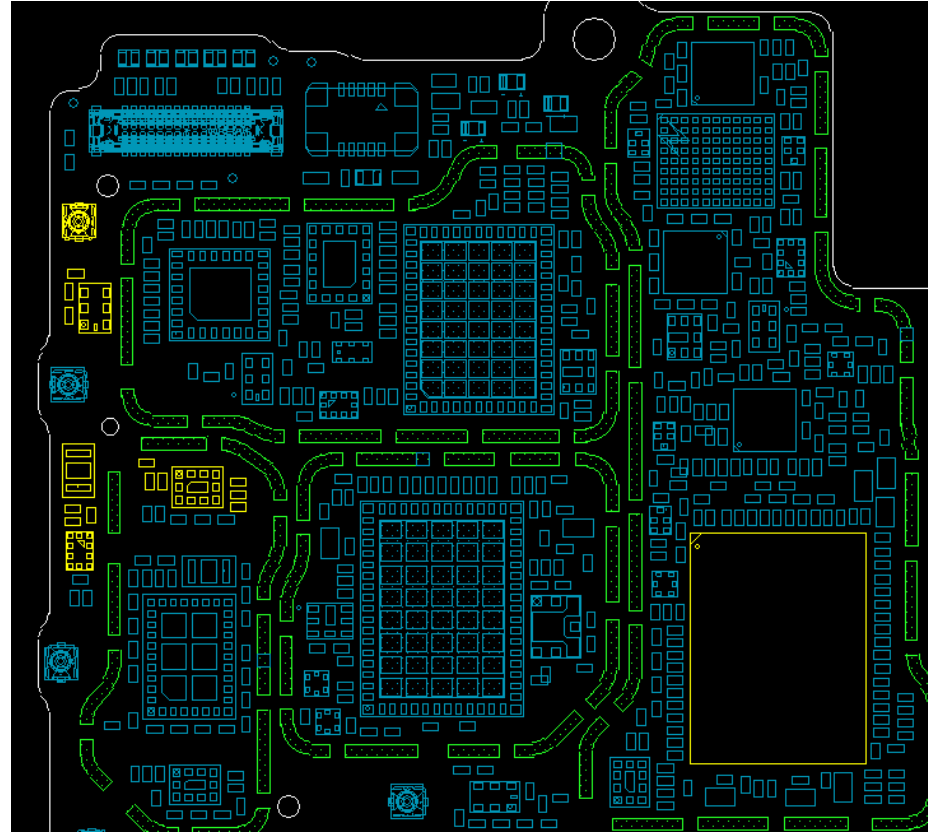
bottom

Div. RF
connector MHB
CON5205

MHB di-plx
U5203

DRx N77 FILTER
U5302

N77 DPDT
U4702



DRx QDM5677
U5301

SDR888
U4101

B1/B3/B7 Tx1 Path

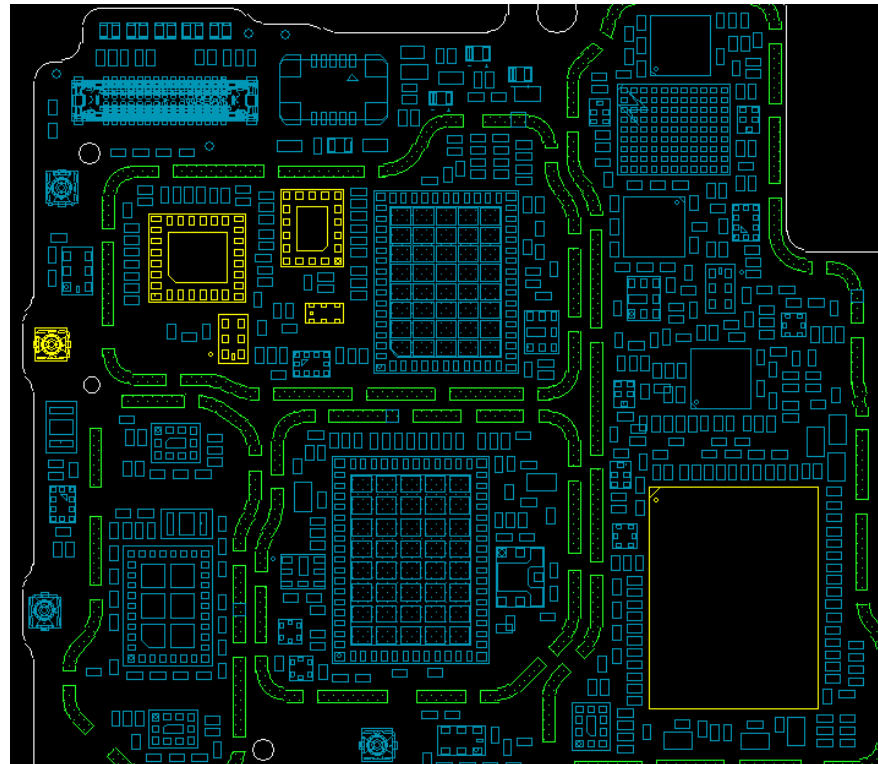
bottom

ENDC filter
U5109

ENDC MMPA
U5101

RF Connector LB
CON4501

SDR88
U4101



LB + B1/B3/B7 DiPX
U4503

ENDC coupler
U5103

B1/B3/B7 PRx MIMO Path

bottom

ENDC filter

U5109

ENDC MMPA

U5101

RF Connector LB

CON4501

QLN5040

U5004

SDR88

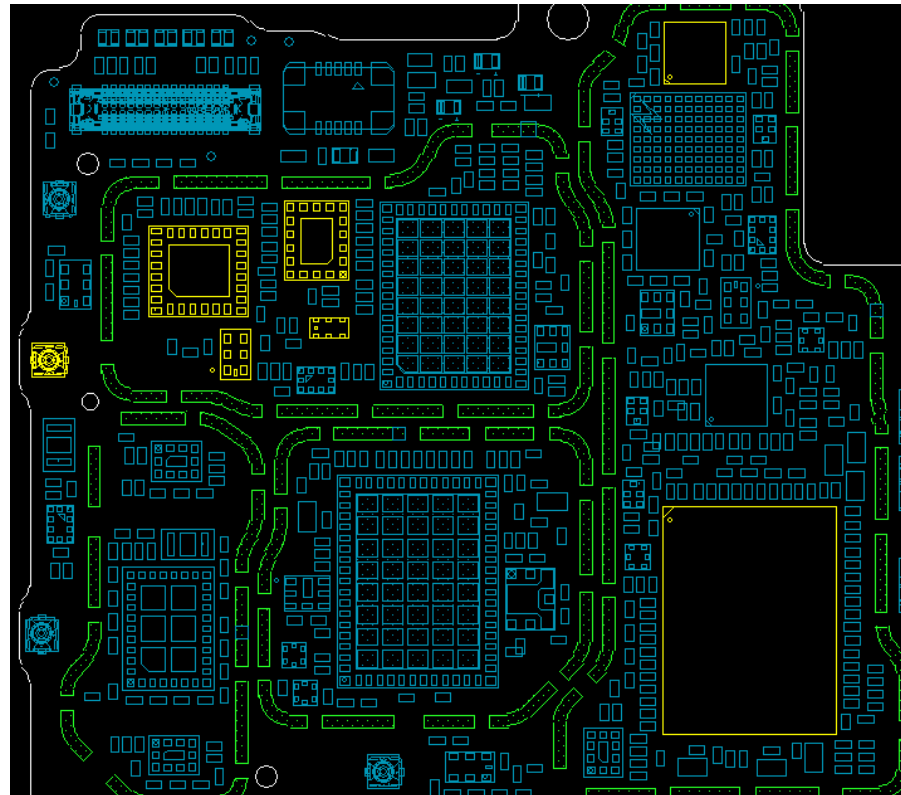
U4101

LB + B1/B3/B7 DiPX

U4503

ENDC coupler

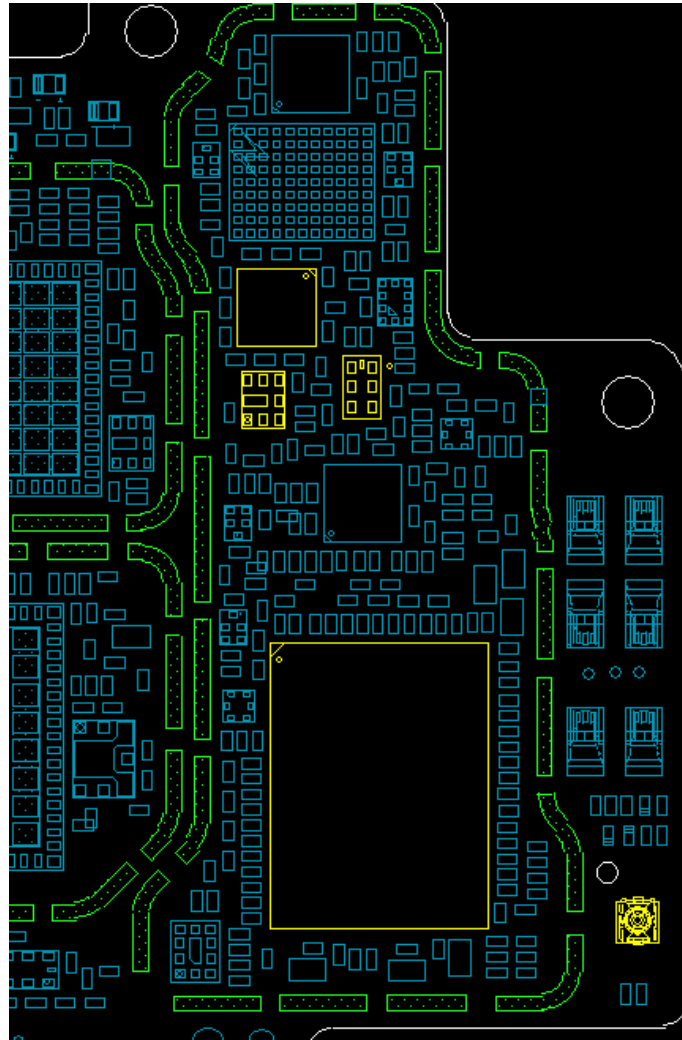
U5103



B1/B3/B7 DRx MIMO Path

Tri-PLX
U5201

SDR888
U4101

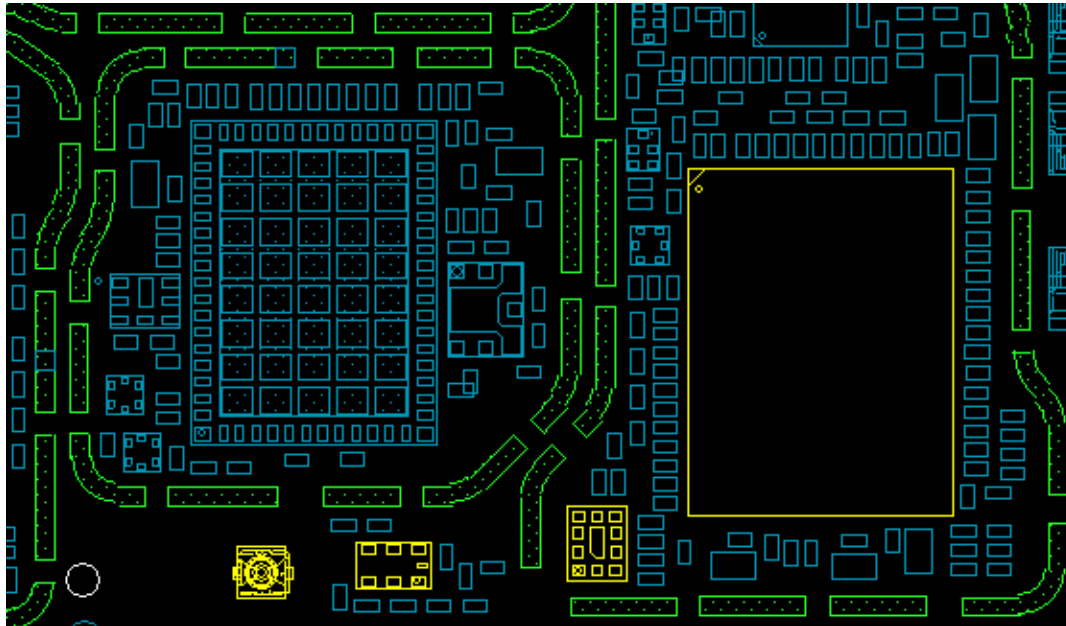


QLN5040
U5005

LB+B1B3B7 DiPX
U5202

Div. RF connector LB
CON5204

B42/N77/N78 PRx MIMO Path



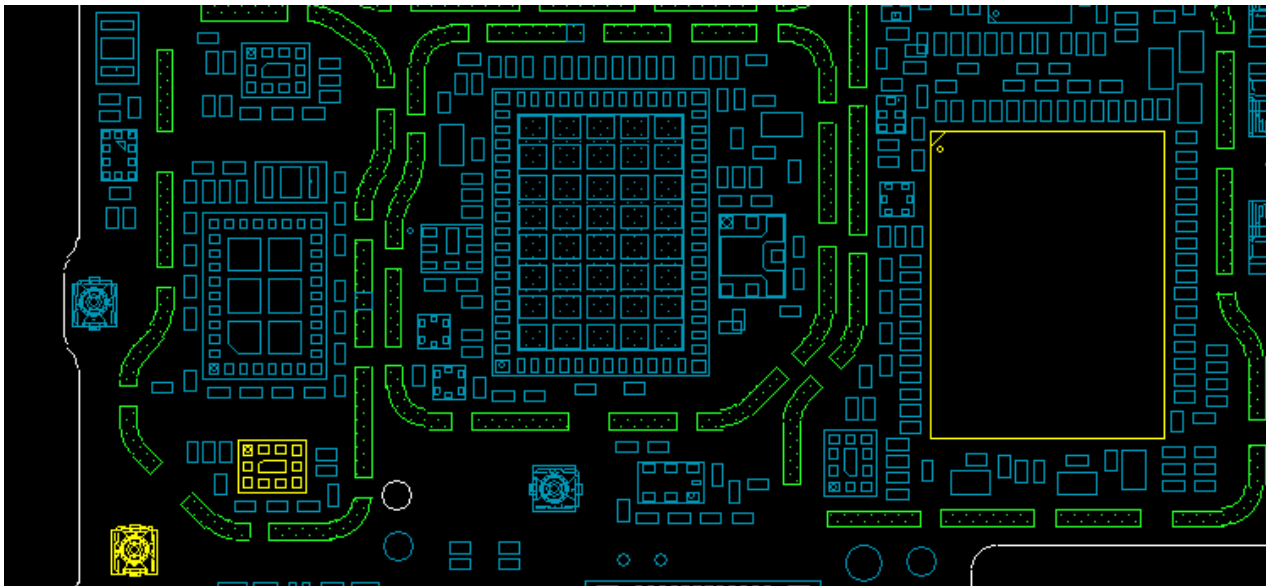
SDR888
U4101

RF Connector MHB
CON4602

MHB + N77 DiPX
U4610

QDM5677
U4604

B42/N77/N78 DRx MIMO Path

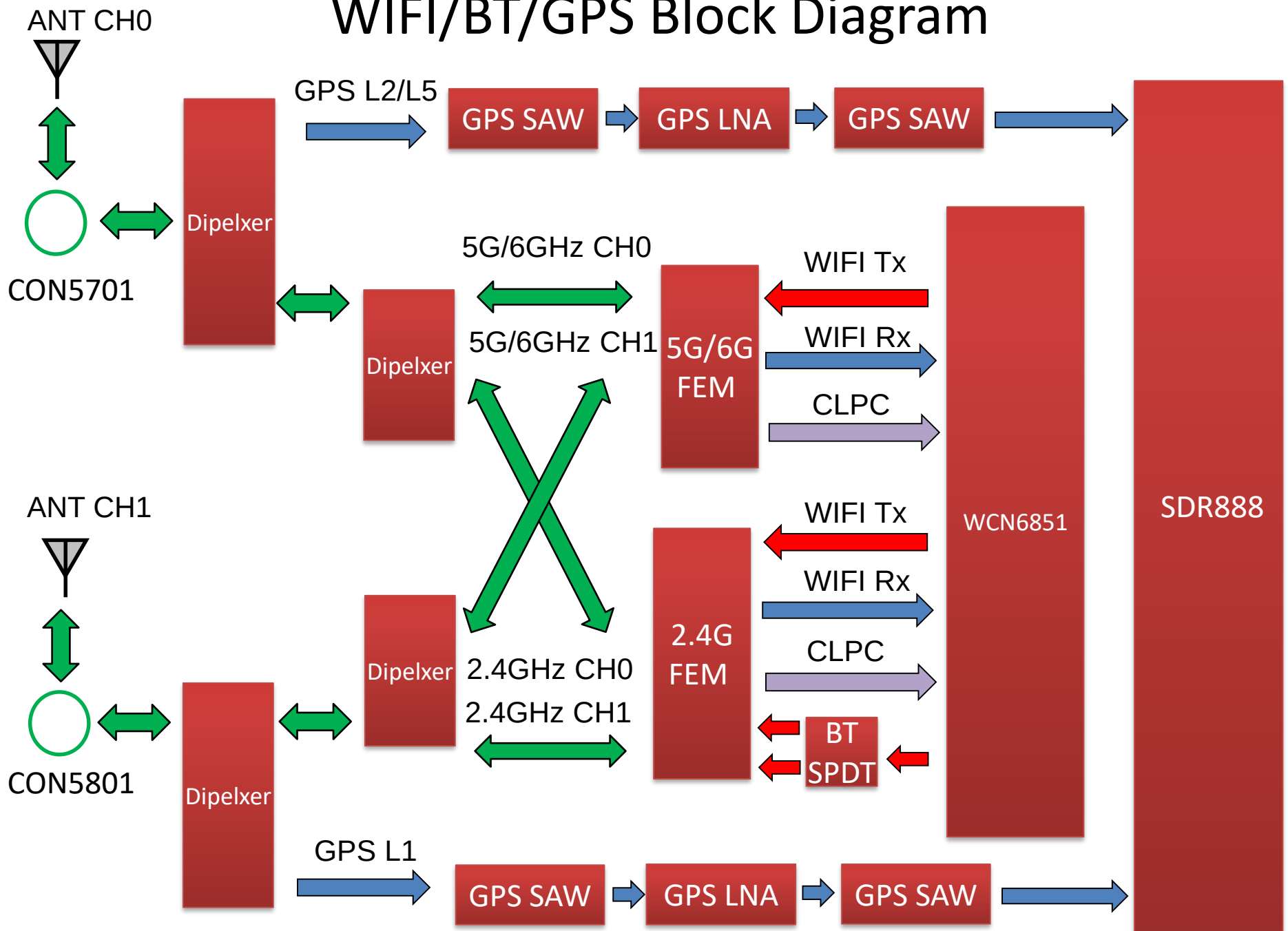


SDR888
U4101

RF Connector MHB
CON5301

QDM5677
U5304

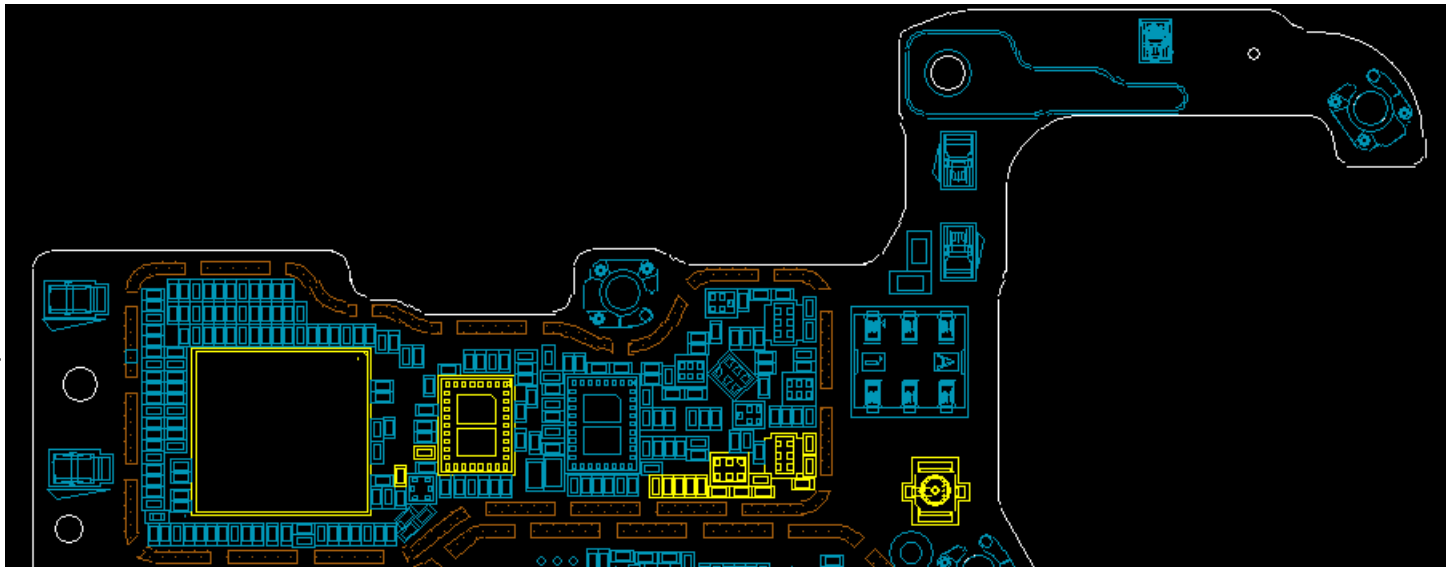
WIFI/BT/GPS Block Diagram



WIFI CH0 2.4G Tx

Top (MB)

WCN6851
U5501



2.4G FEM
U5705

CH0 DiPX
U5703

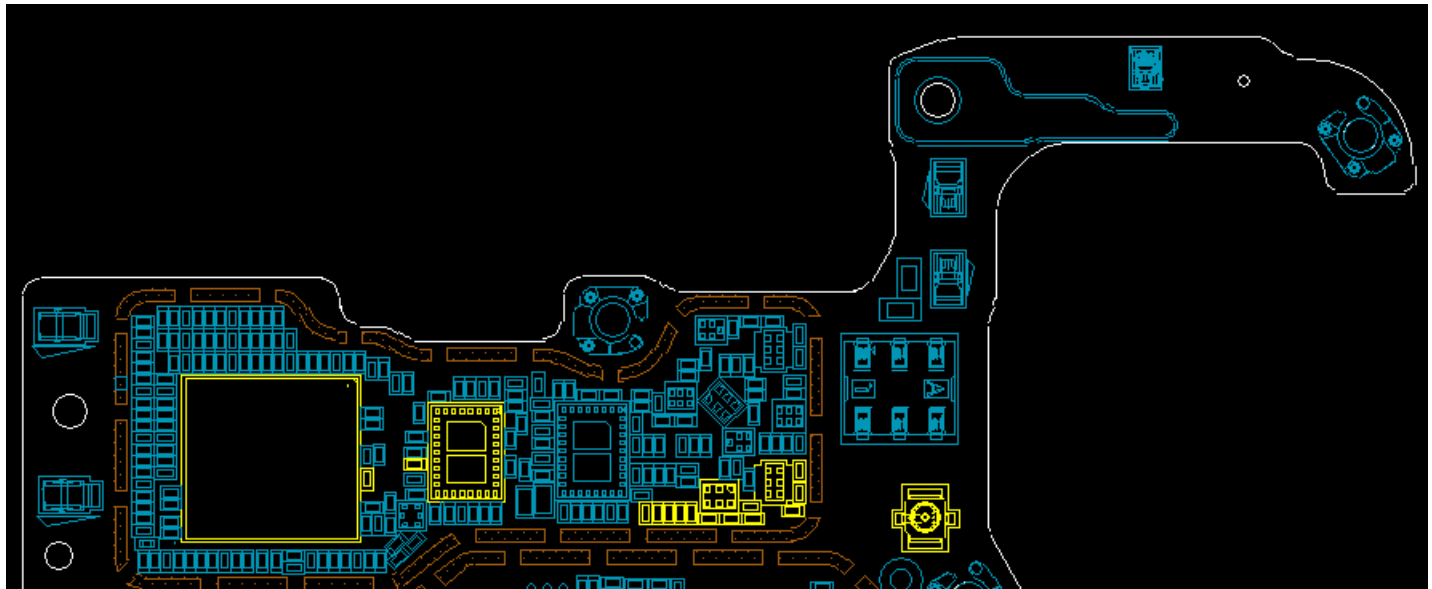
CH0 DiPX 2.4/5
U5708

CH0 Connector
CON5701

WIFI CH0 2.4G Rx

Top (MB)

WCN6851
U5501



2.4G FEM
U5705

CH0 DiPX
U5703

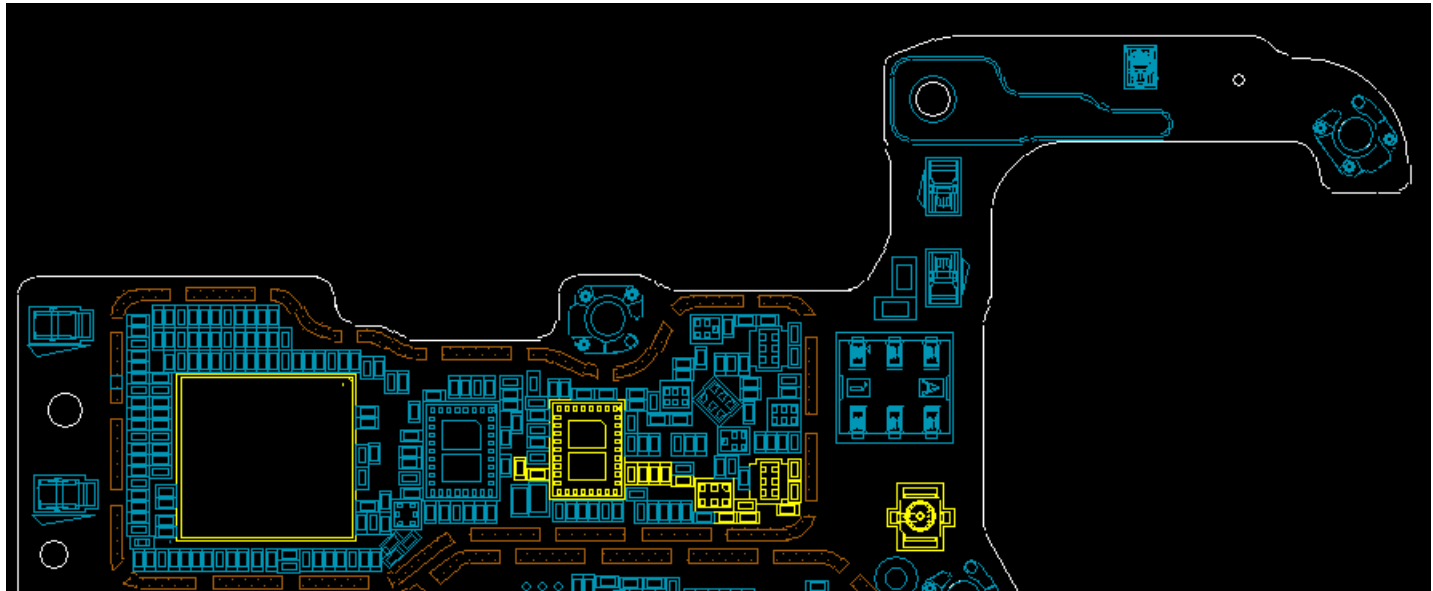
CH0 Connector
CON5701

CH0 DiPX 2.4/5
U5708

WIFI CH0 5G Tx

Top (MB)

WCN6851
U5501



5G FEM
U5809

CH0 DiPX
U5703

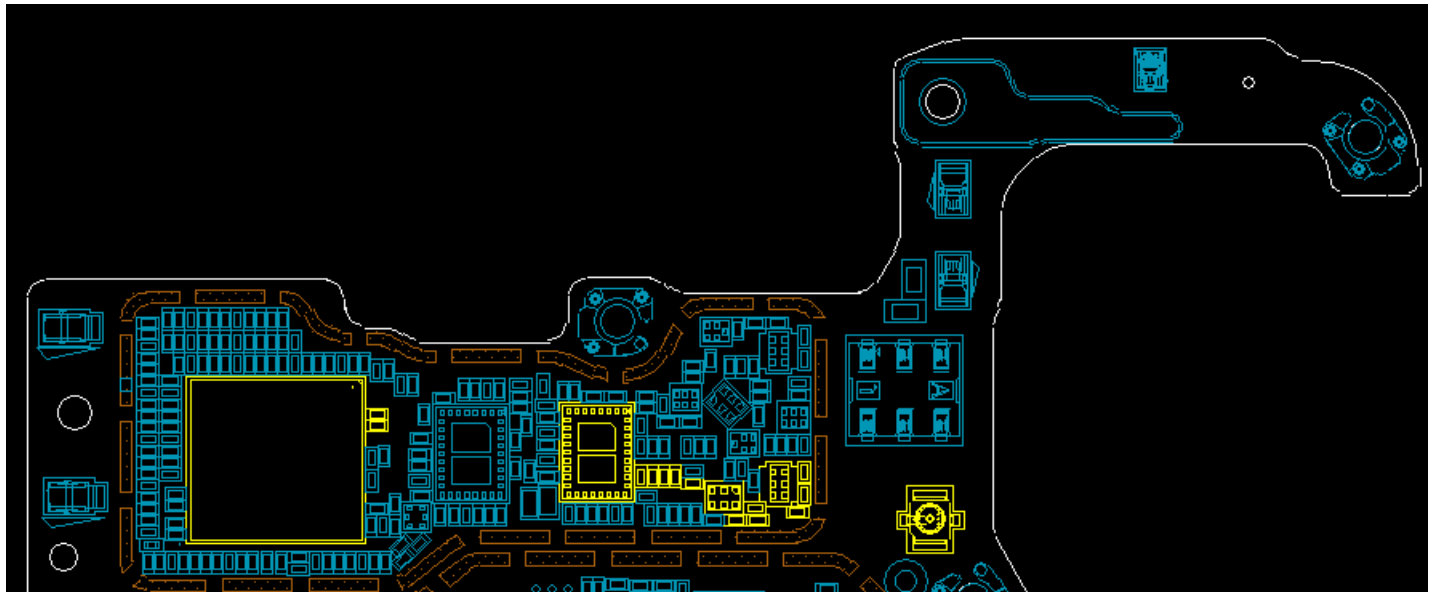
CH0 DiPX 2.4/5
U5708

CH0 Connector
CON5701

WIFI CH0 5G Rx

Top (MB)

WCN6851
U5501



5G FEM
U5809

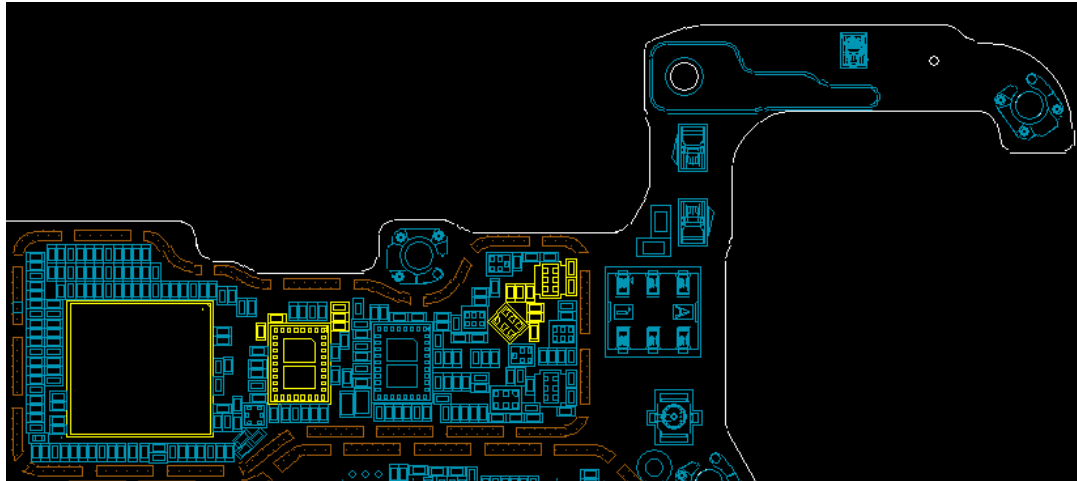
CH0 DiPX
U5703

CH0 DiPX 2.4/5
U5708

CH0 Connector
CON5701

WIFI CH1 2.4G Tx

Top (MB)



Bottom (MB)



CH1 Connector
CON5801

WCN6851
U5501

2.4G FEM
U5705

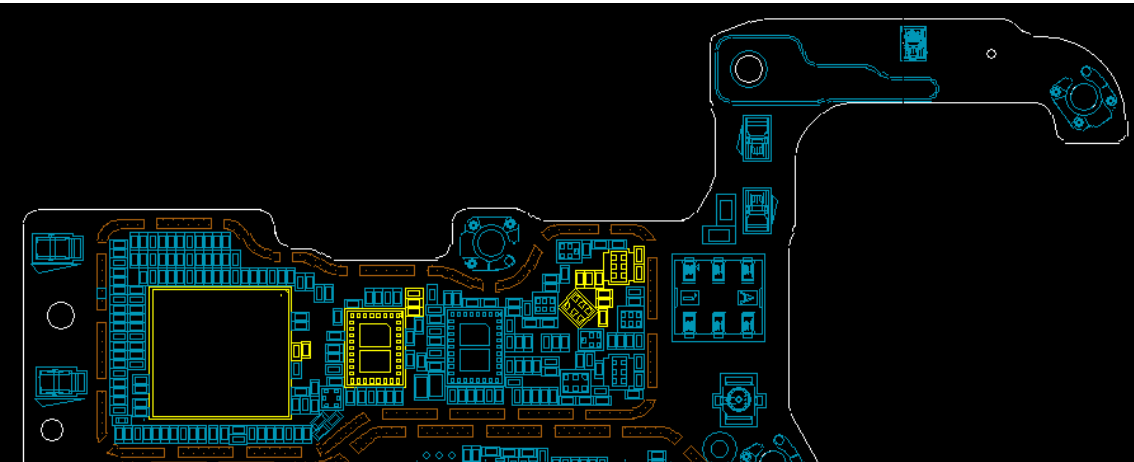
CH1 DiPX
U5811

CH1 DiPX 2.4/5
U5810

WIFI CH1 2.4G Rx

Top (MB)

Bottom (MB)



CH1 Connector
CON5801

WCN6851
U5501

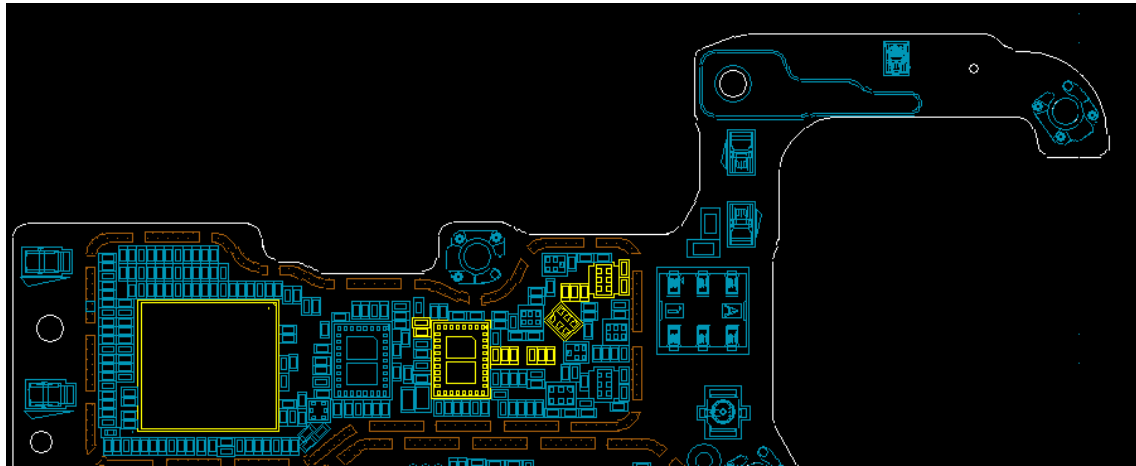
2.4G FEM
U5705

CH1 DiPX
U5811

CH1 DiPX 2.4/5
U5810

WIFI CH1 5G Tx

Top (MB)



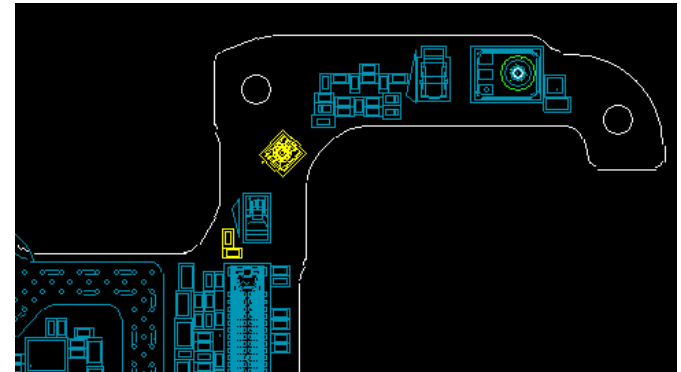
WCN6851
U5501

5G FEM
U5809

CH1 DiPX
U5811

CH1 DiPX 2.4/5
U5810

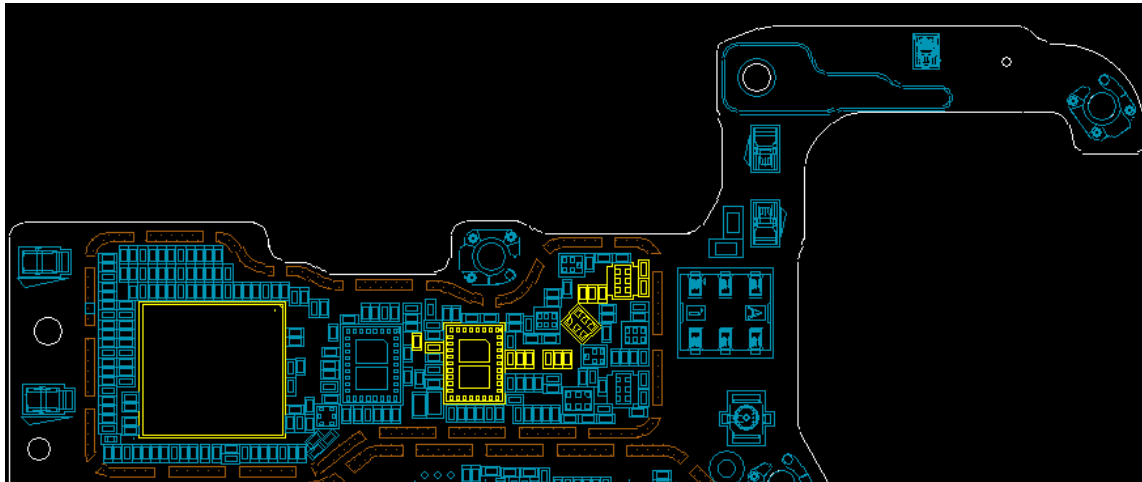
Bottom (MB)



CH1 Connector
CON5801

WIFI CH1 5G Rx

Top (MB)



WCN6851
U5501

5G FEM
U5809

CH1 DiPX
U5811

CH1 DiPX 2.4/5
U5810

Bottom (MB)



CH1 Connector
CON5801

BT CH0 Tx

Top (MB)

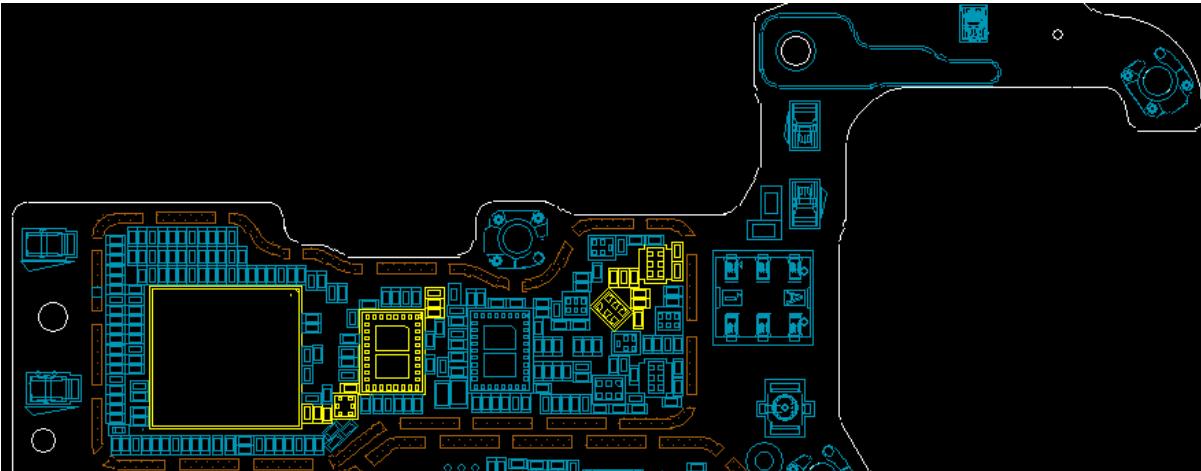
WCN6851
U5501



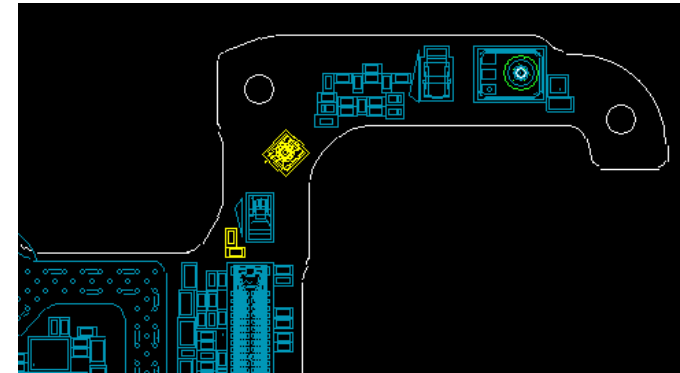
BT SPDT U5707	2.4G FEM U5705	CH0 DiPX U5703	CH0 Connector CON5701
	CH0 DiPX 2.4/5 U5708		

BT CH1 Tx

Top (MB)



Bottom (MB)



CH1 Connector

CON5801

WCN6851

U5501

2.4G FEM

U5705

CH1 DiPX

U5811

BT SPDT

U5707

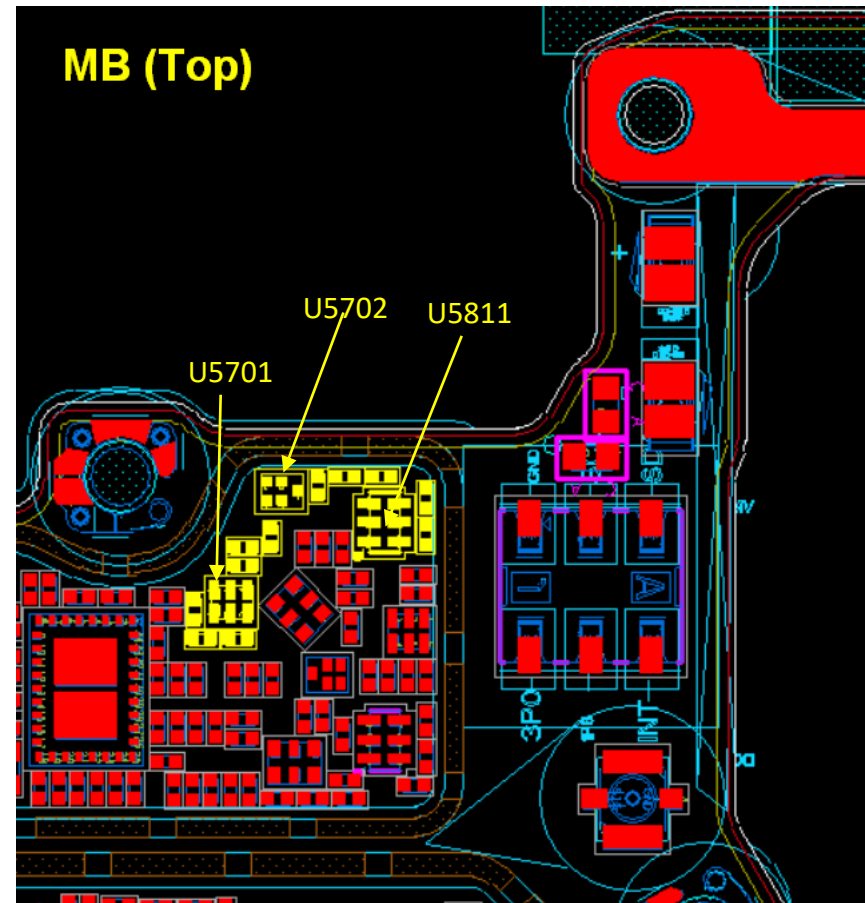
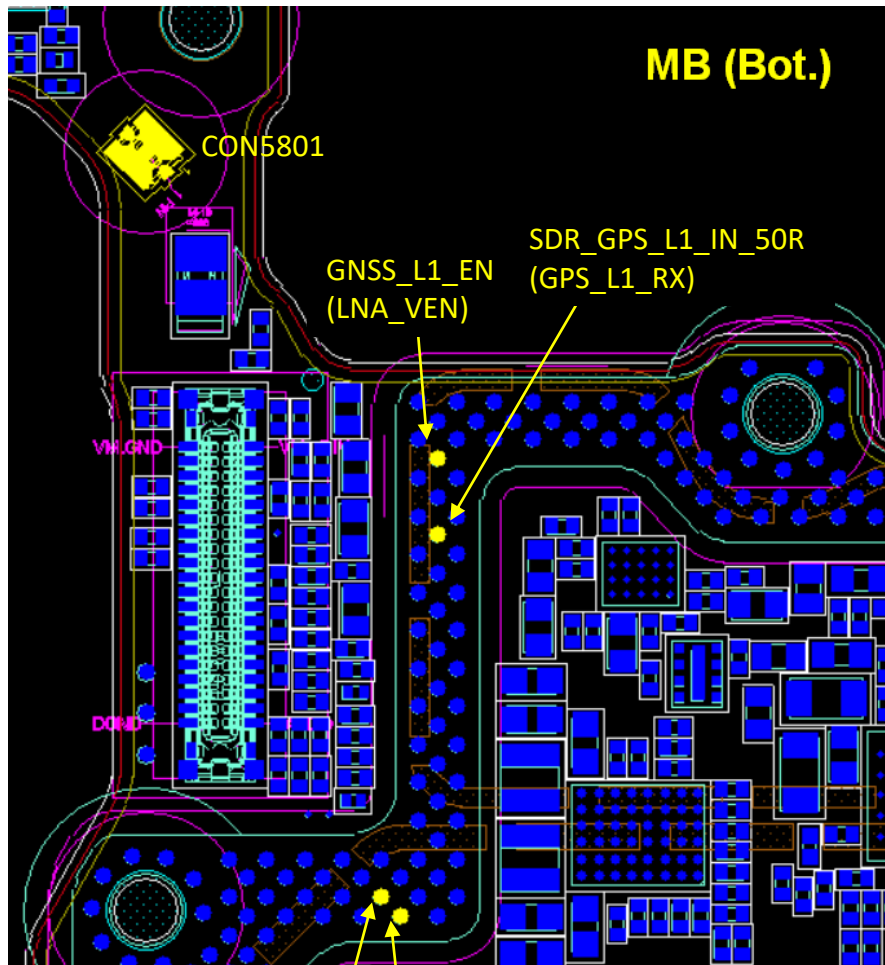
CH1 DiPX 2.4/5

U5810

ZS590KS Trouble Shooting Guide

GPS/NFC/FM

GPS_L1



GPS_L1 path

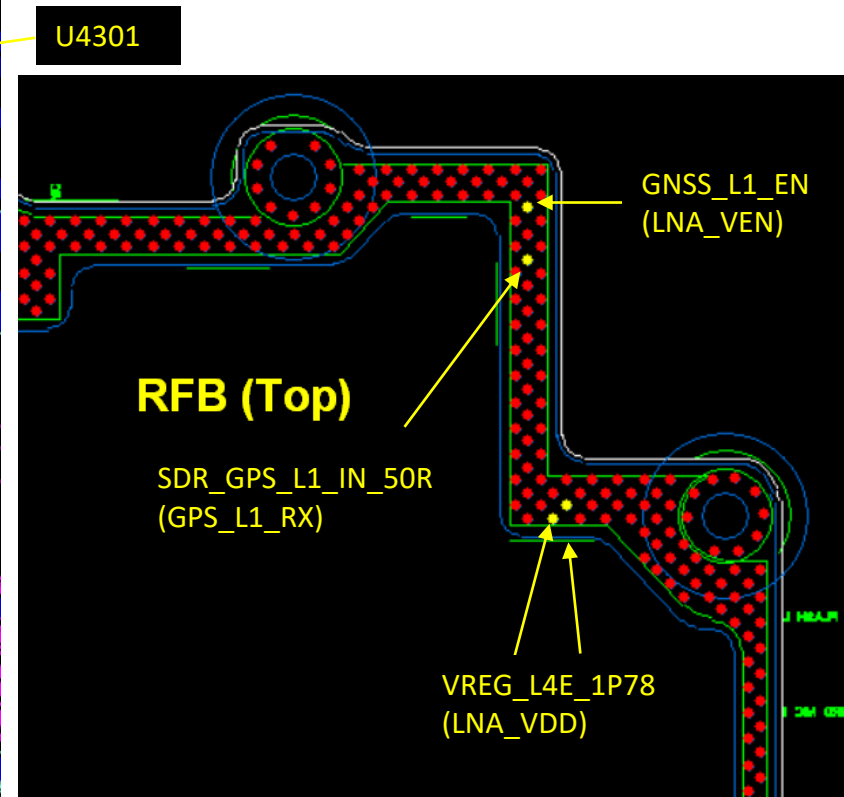
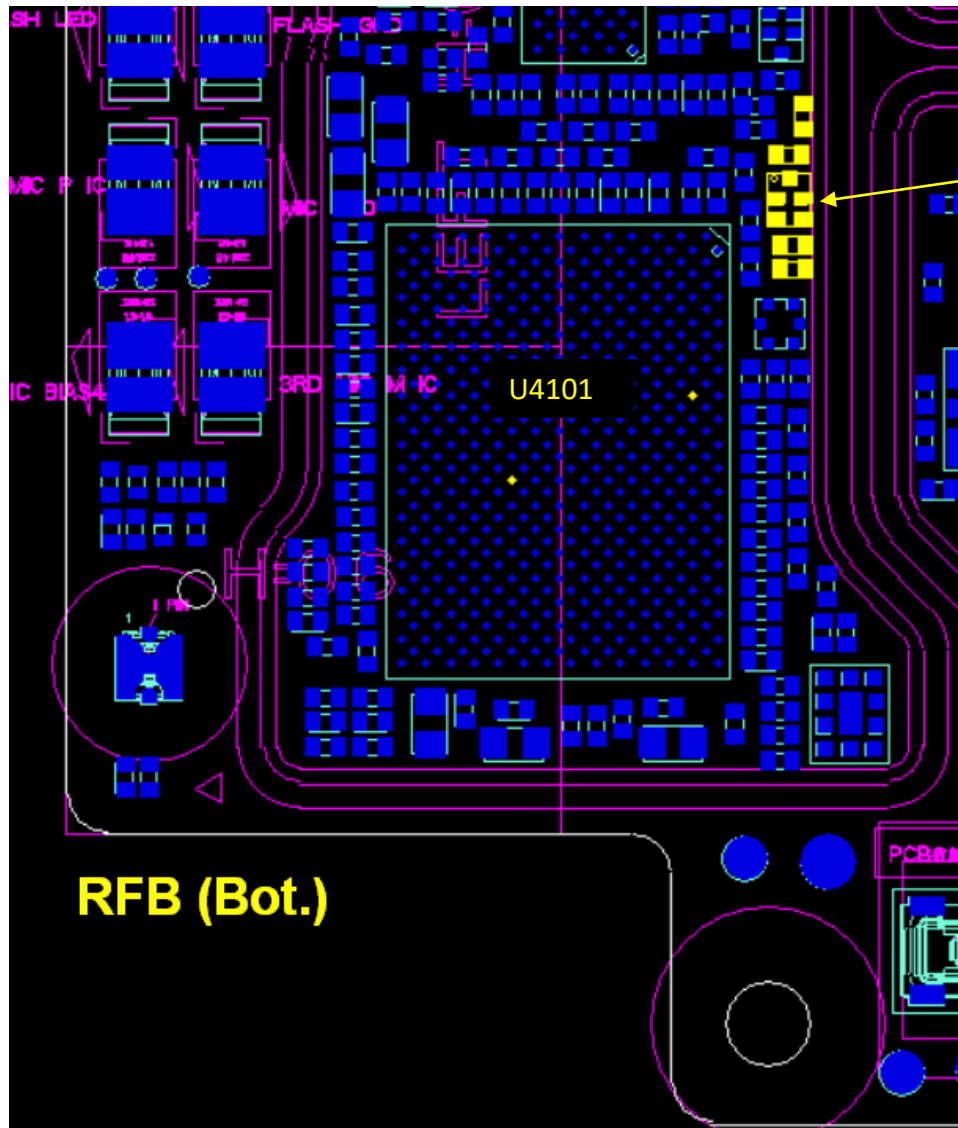
CON5801: RF SWITCH CONNECTOR

U5811: TRIPLEXER

U5702: Pre SAW

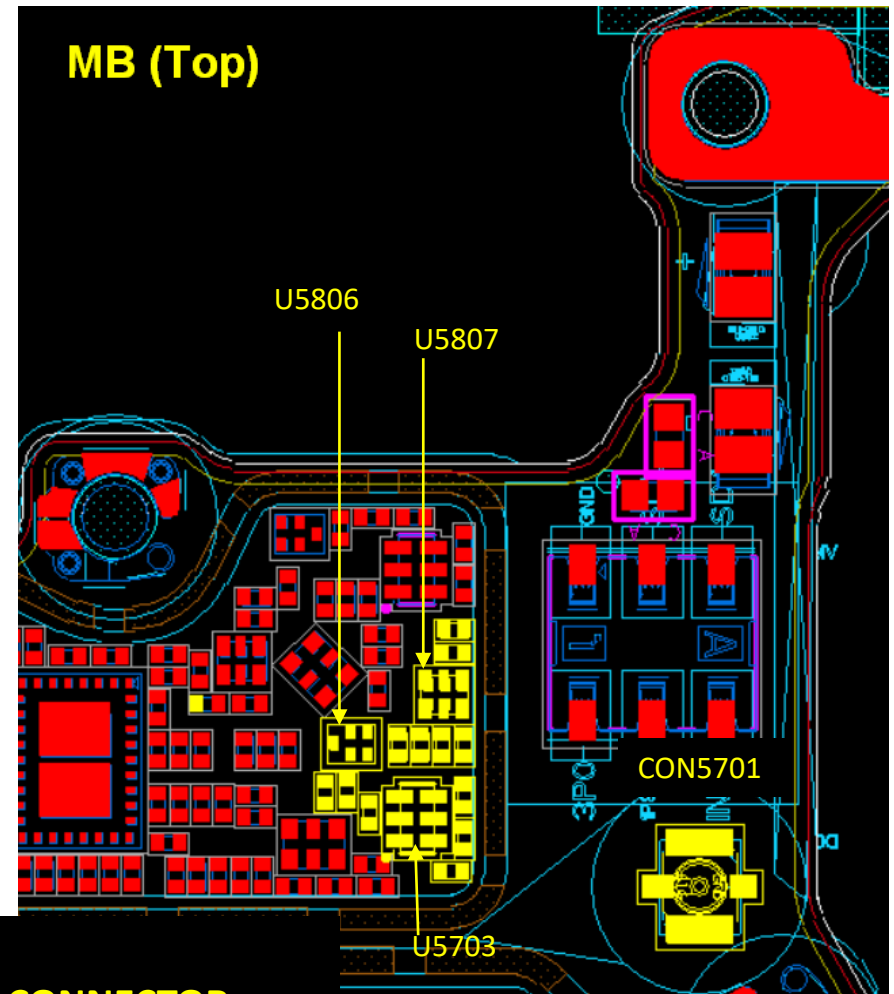
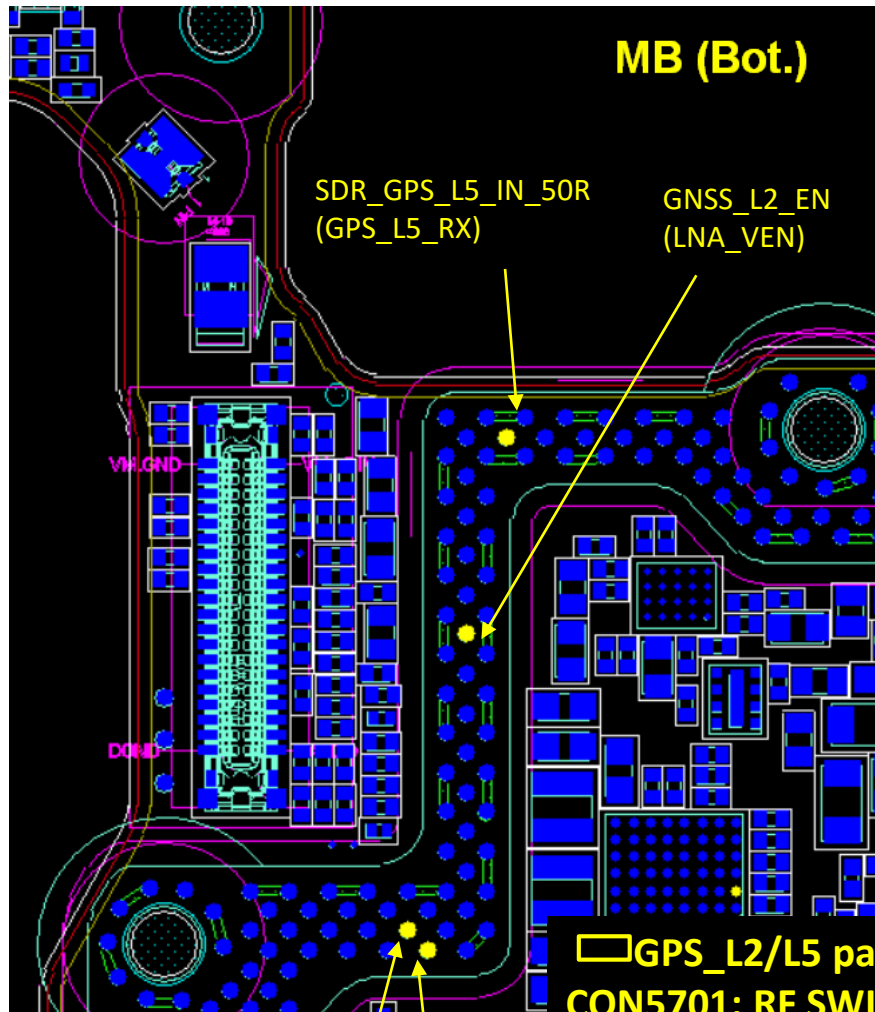
U5701: LNA

GPS_L1



U4301: Post SAW
U4101: Transceiver/SDR868

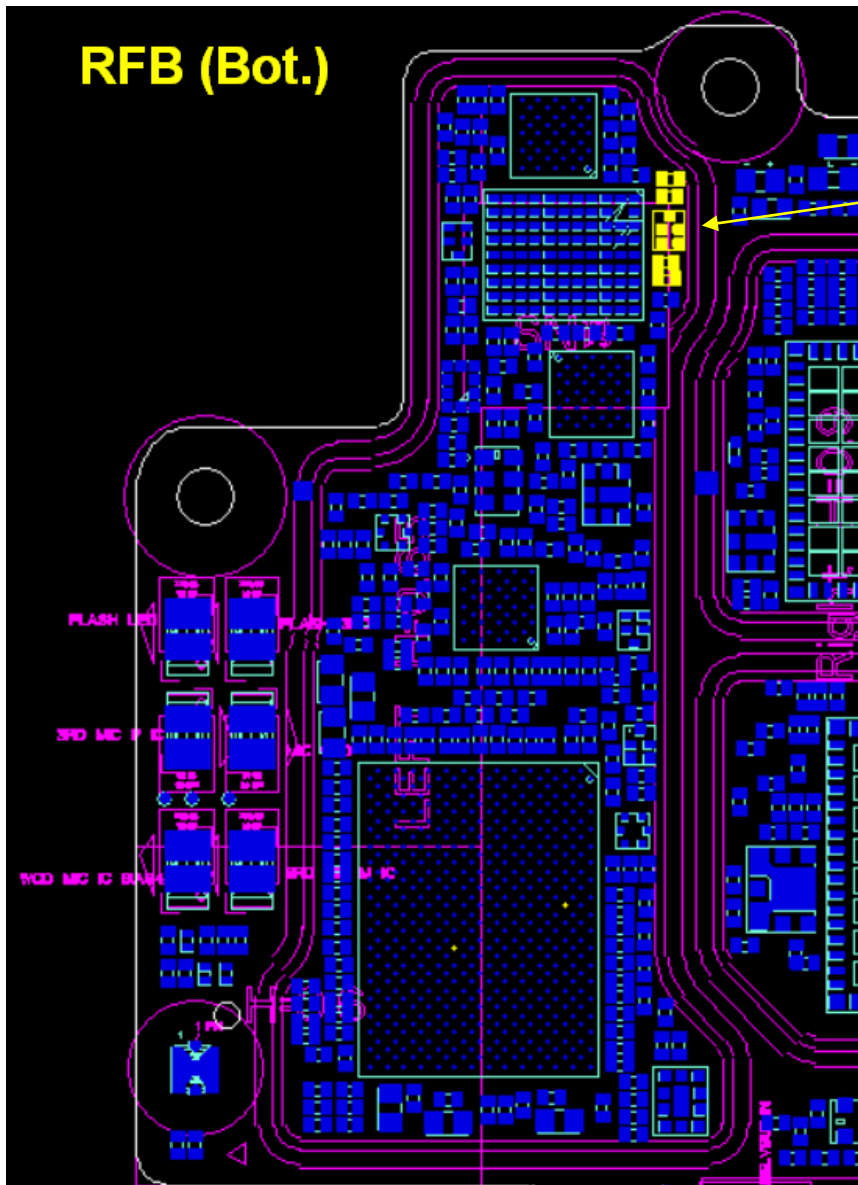
GPS_L2/L5



□ GPS_L2/L5 path
CON5701: RF SWITCH CONNECTOR
U5703: TRIPLEXER
U5806: Pre SAW
U5807: LNA

GPS_L2/L5

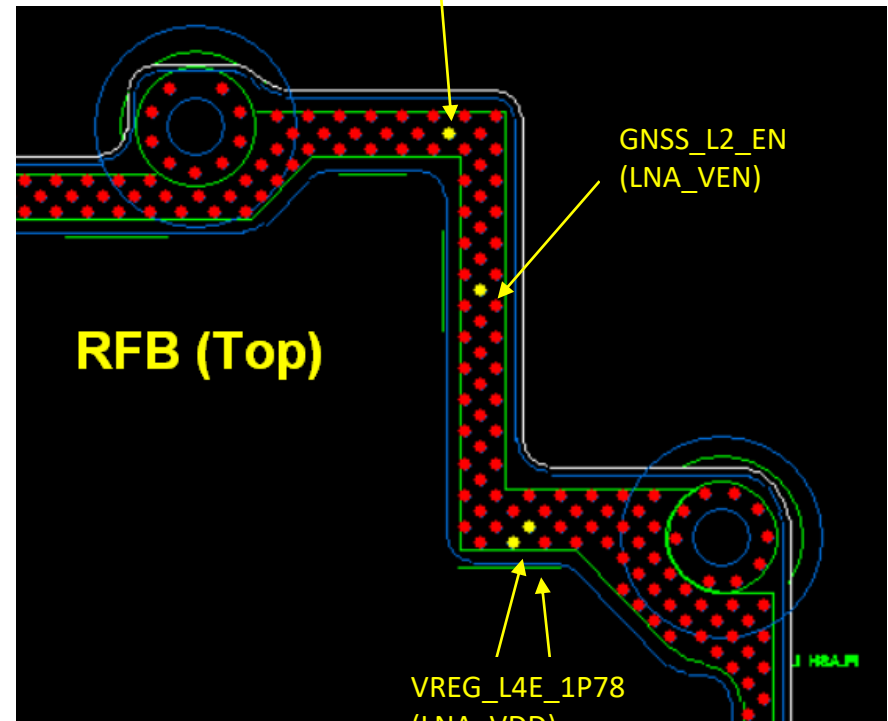
RFB (Bot.)



U4302

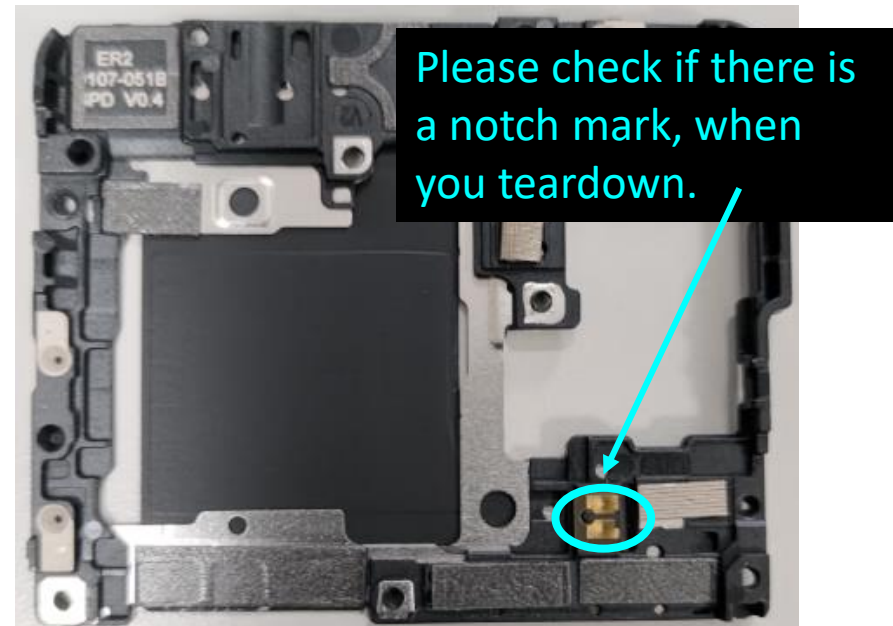
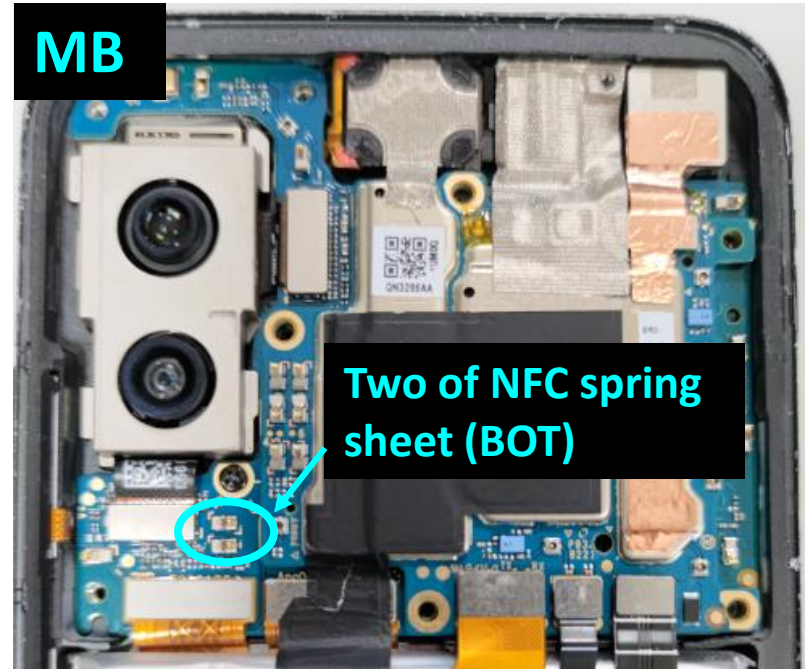
SDR_GPS_L5_IN_50R
(GPS_L5_RX)

RFB (Top)



U4302: Post SAW
U4101: Transceiver/SDR868

NFC



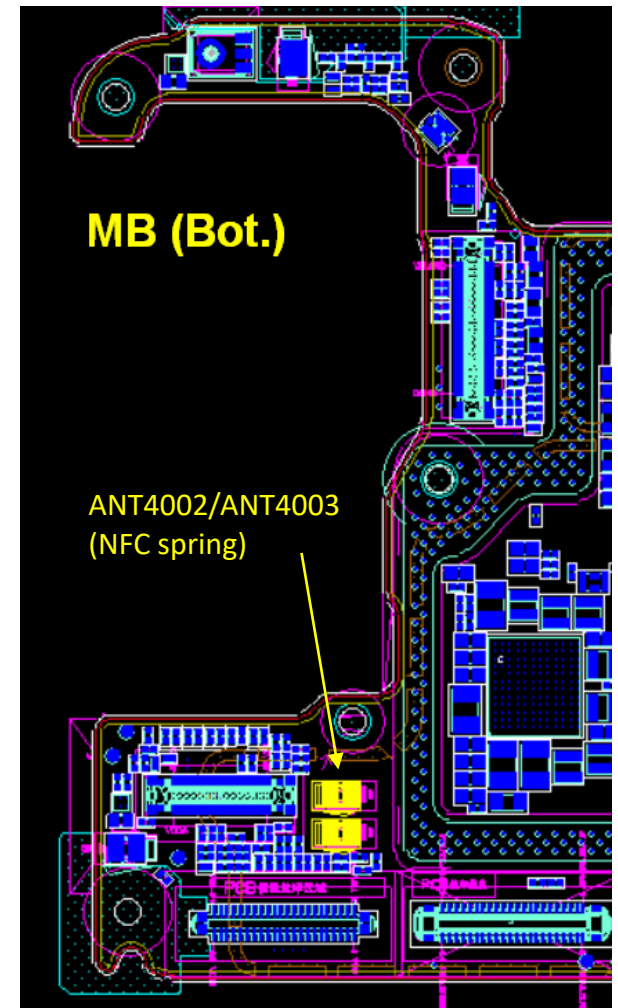
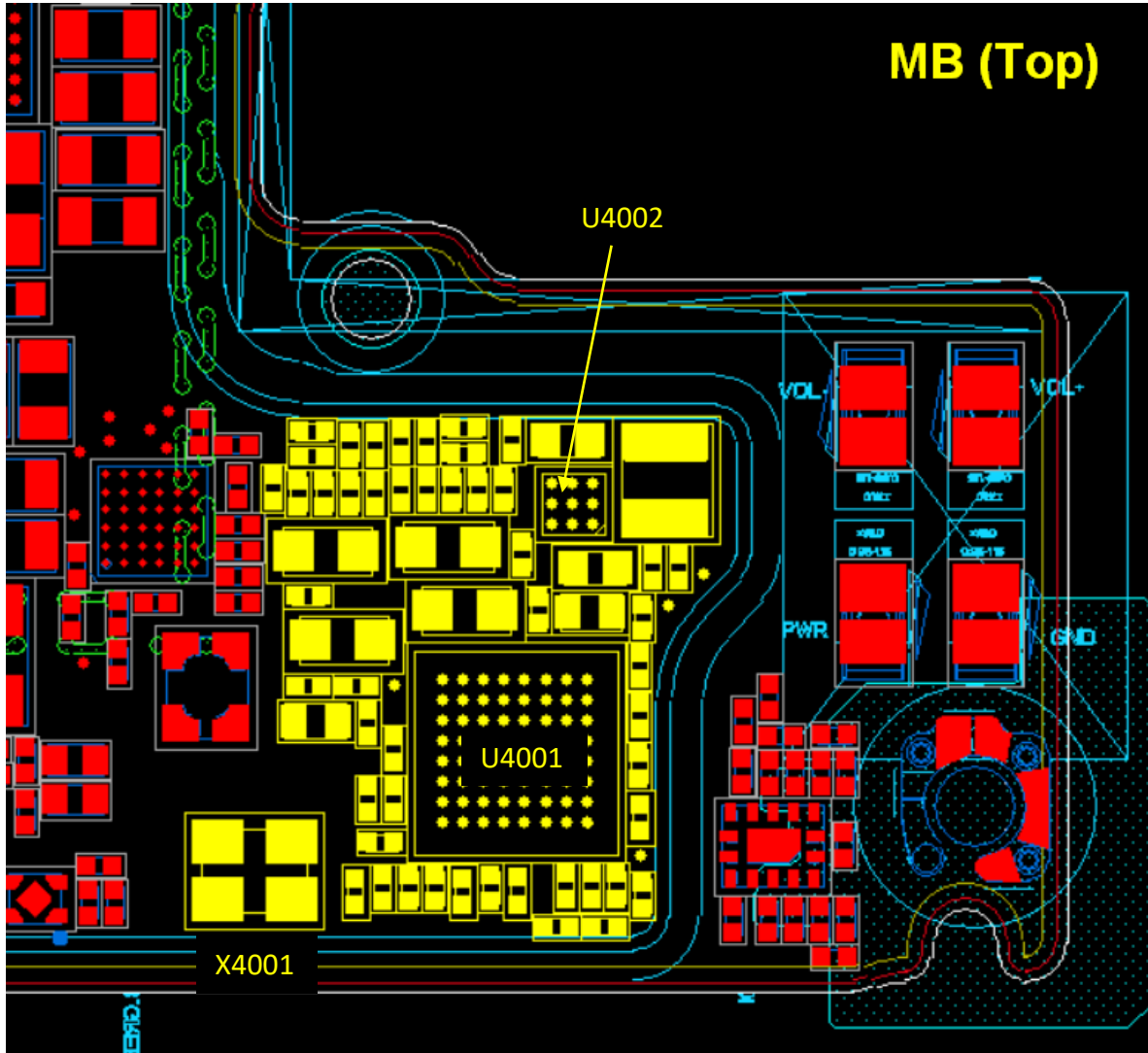
NFC

□ NFC components

U4001: NFC CHIP

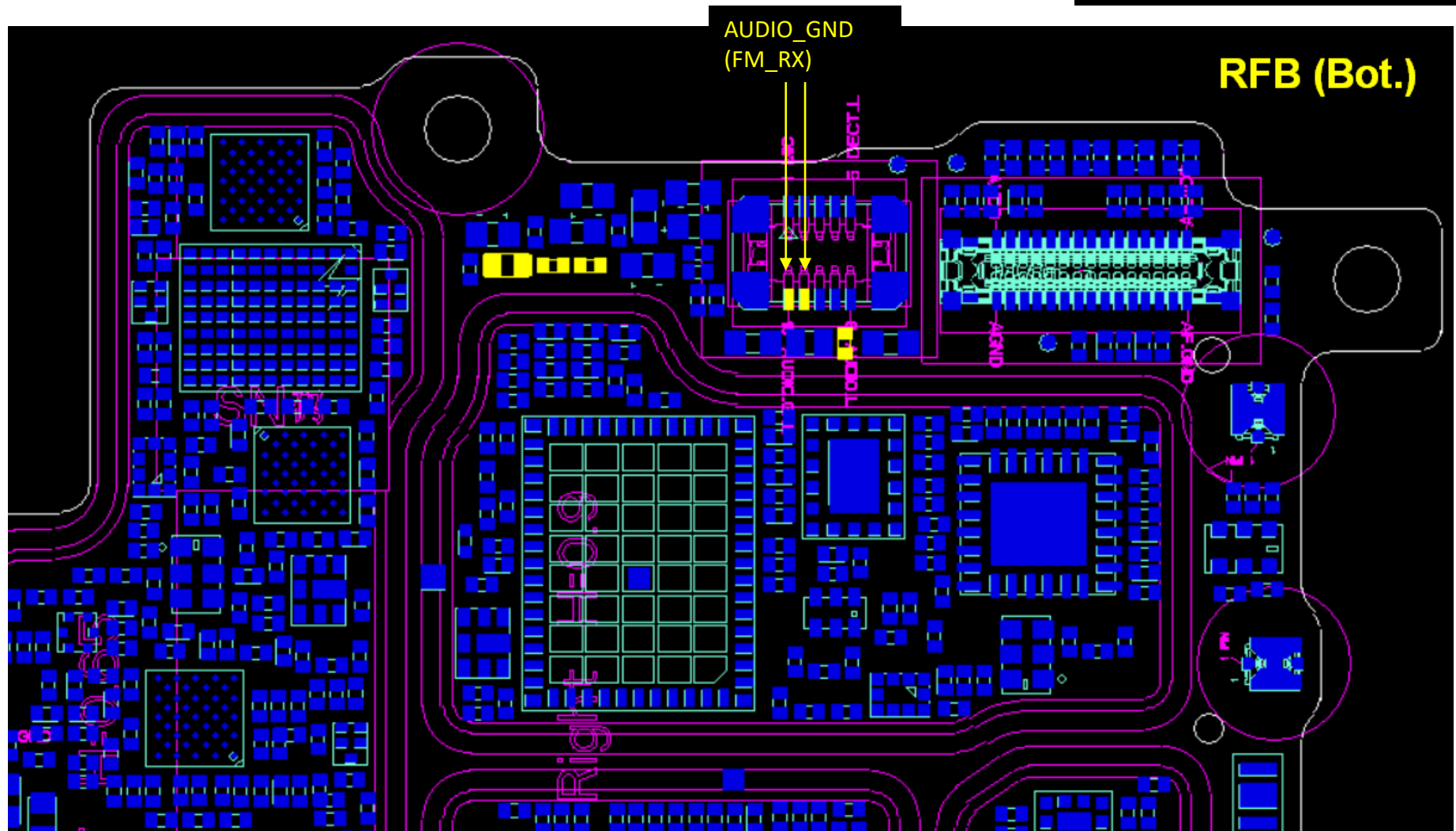
U4002: DC/DC CONVERTER

X4001: XTAL 27.12MHZ



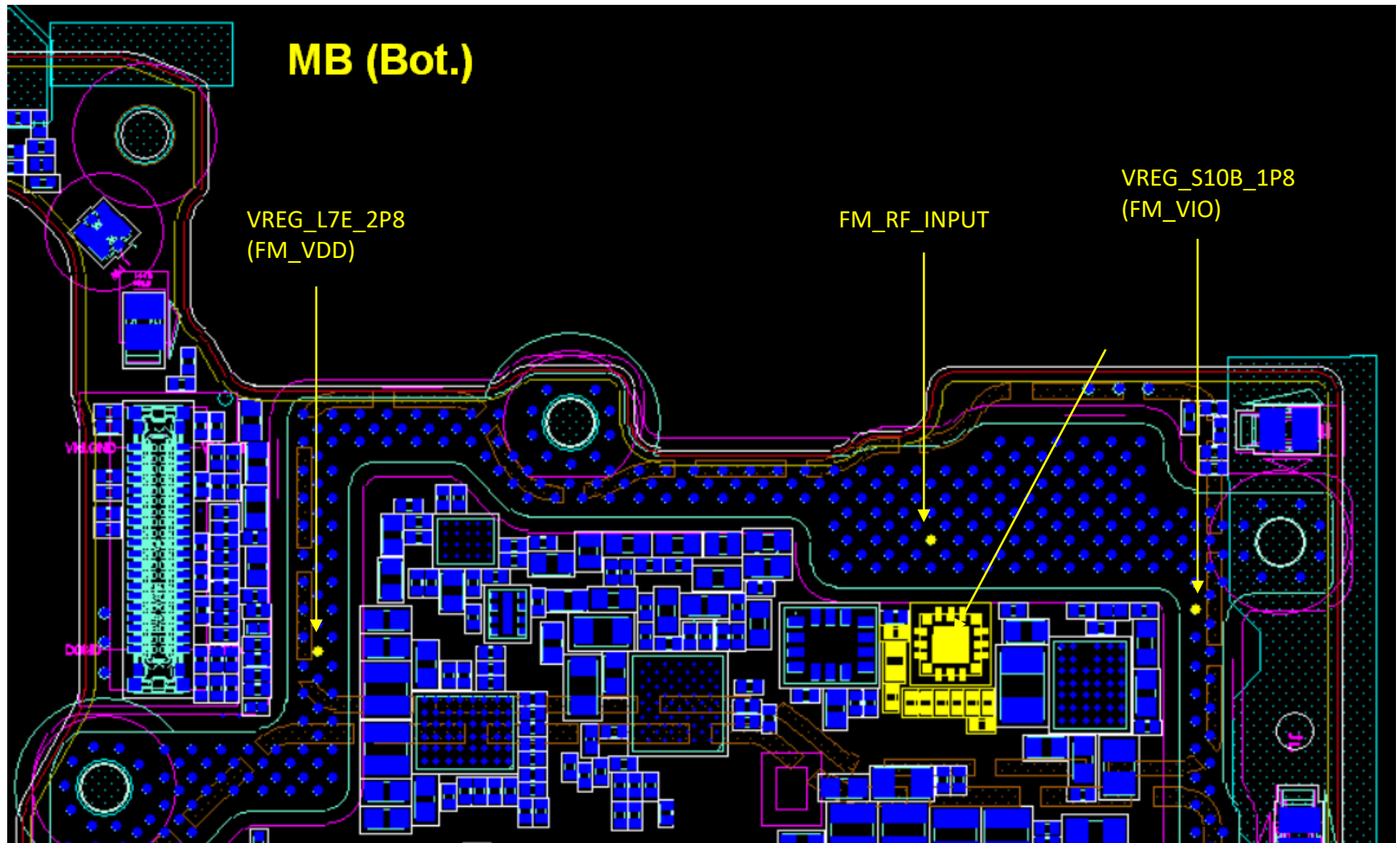
FM

FM components



FM

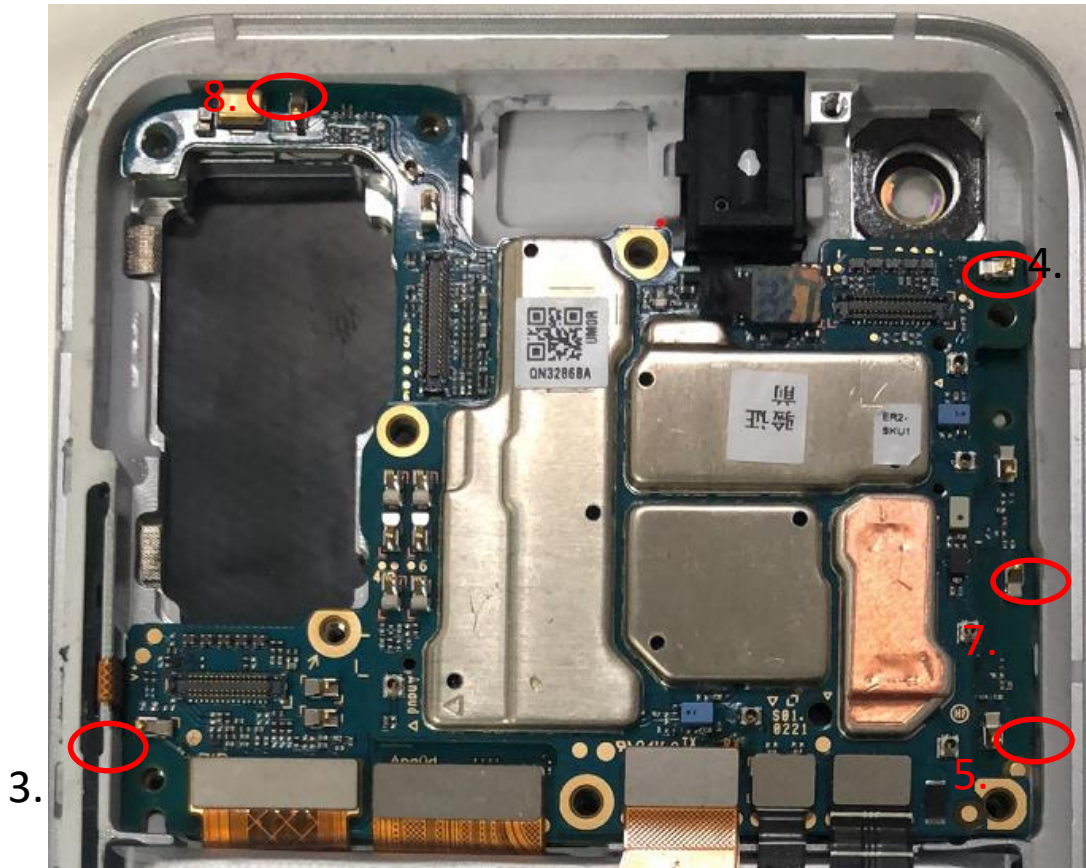
FM components
U3001: FM CHIP



Sake Trouble shooting Guide

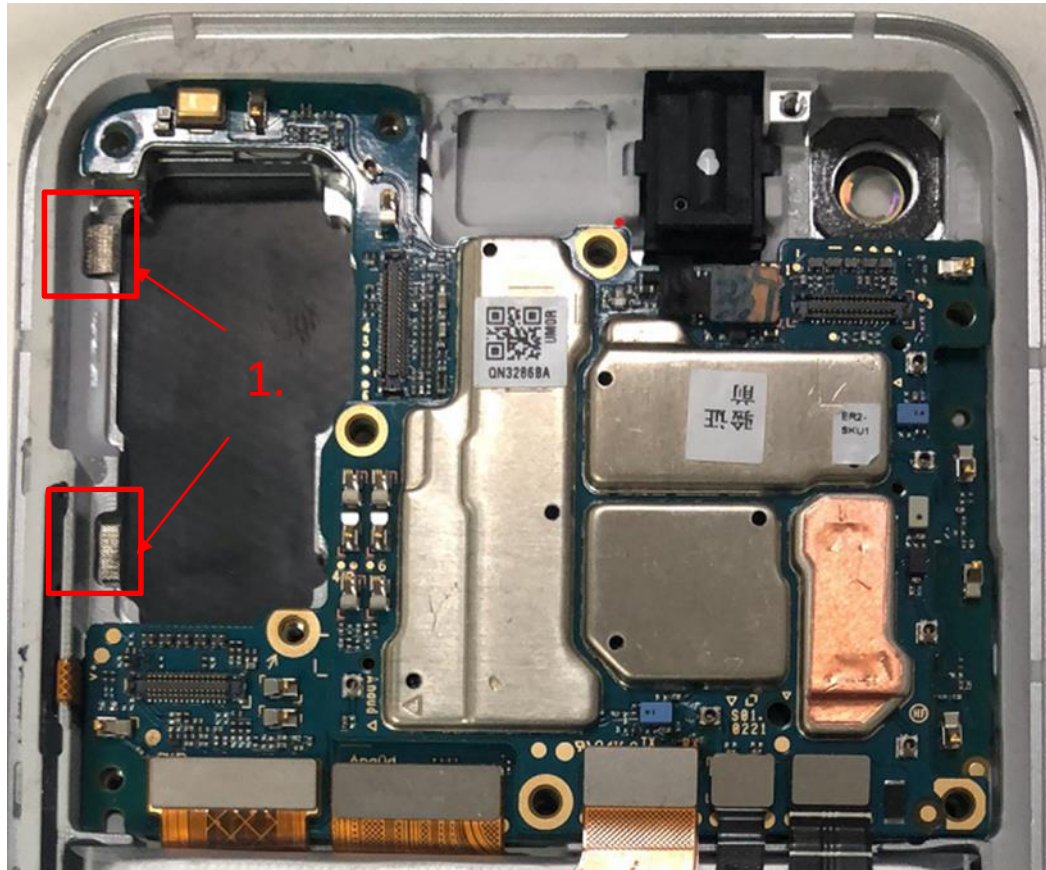
Antenna part

Upper antenna feed point :



- 3. LTE Div. B71/B28/B20/B5/B8+MIMO B3/B1/B7
- 4. LTE Div. B3/B2/B1/7/B38/B39/B40/B41/n77/n78
- 5. NR TRX. n77
- 7. WiFi 2.4 GHz/5GHz Ch0+ GPS L5+WiFi 6e
- 8. WiFi 2.4 GHz/5GHz Ch1+ GPS L1+WiFi 6e

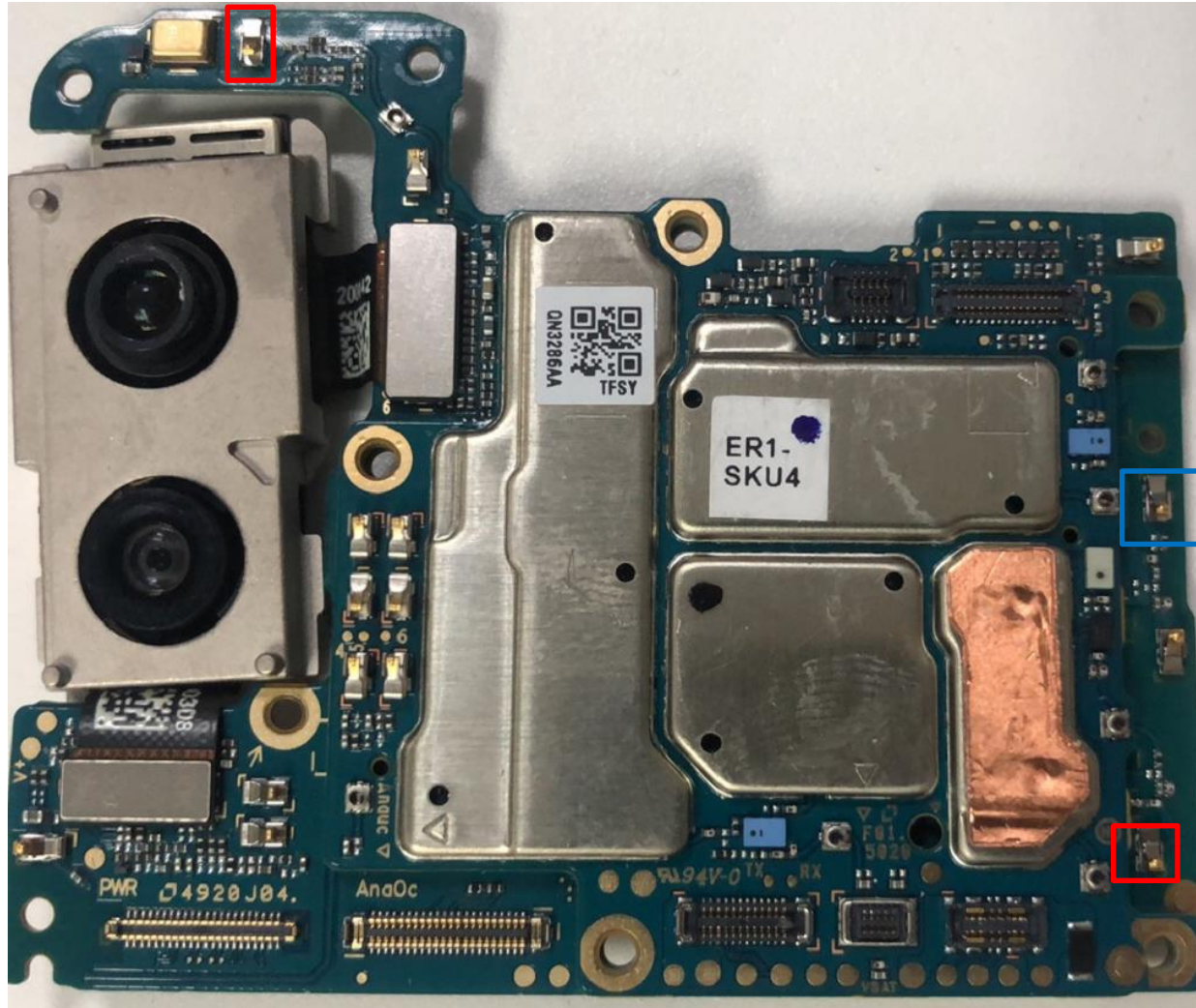
Rear camera grounding



1. WiFi 2.4 GHz/5GHz Ch1+ GPS L1+WiFi 6e

Main board spring sheet: All spring sheets need to avoid deformation/missing parts

Ch1_WIFI6E

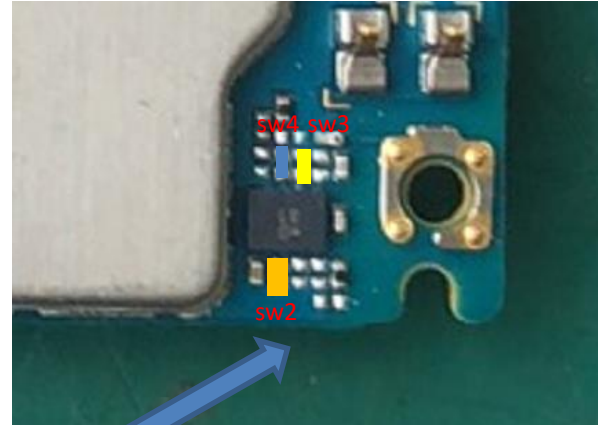
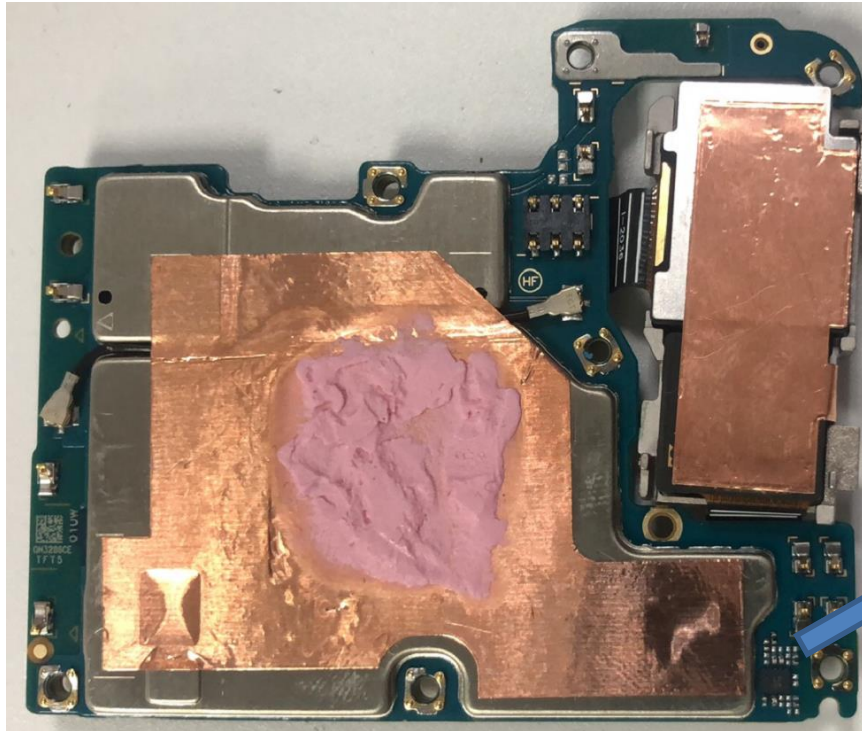


Ch0_WIFI6E

n77(TRX)

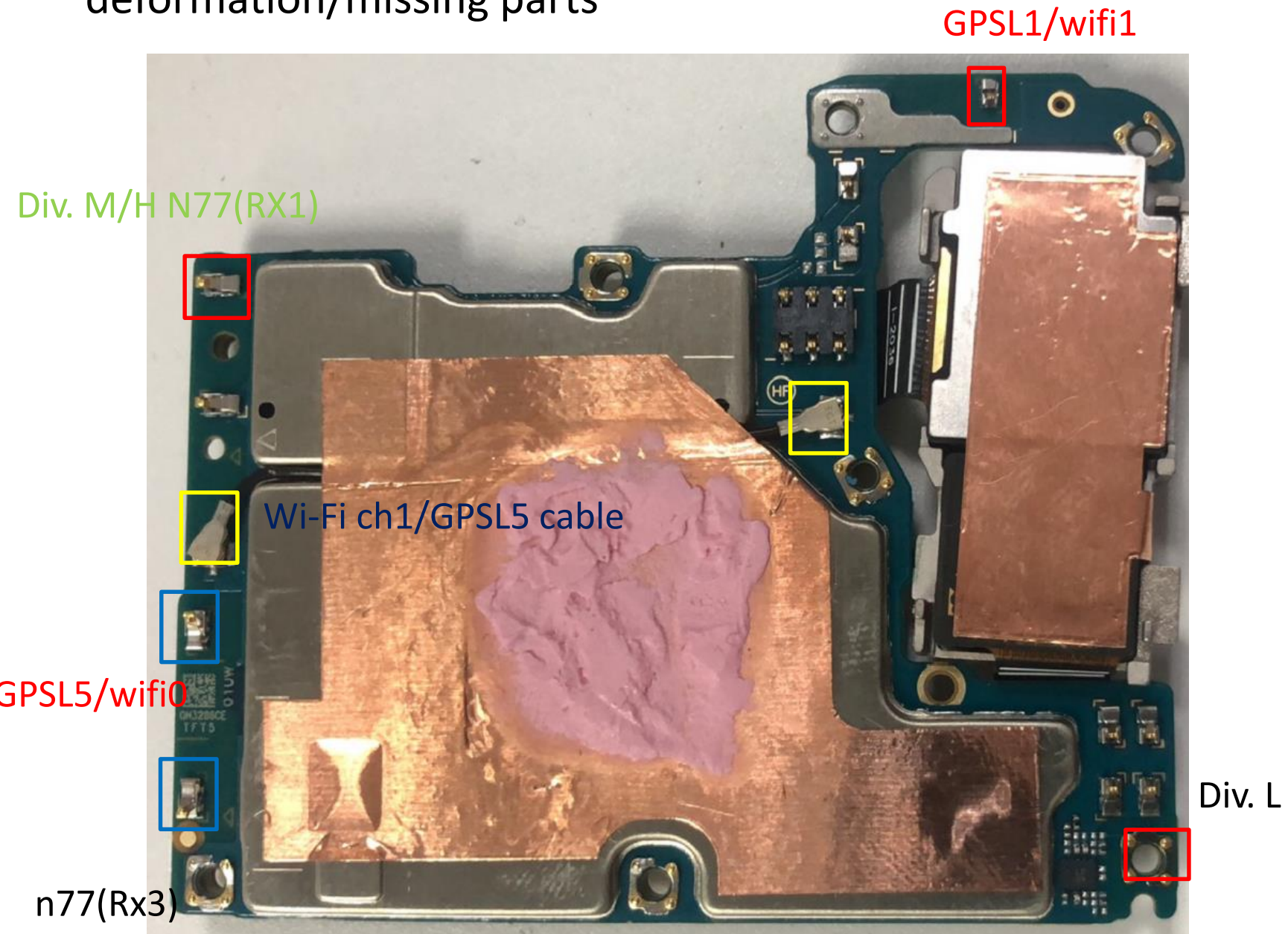
Lower antenna notice:

1. Antenna aperture component location,



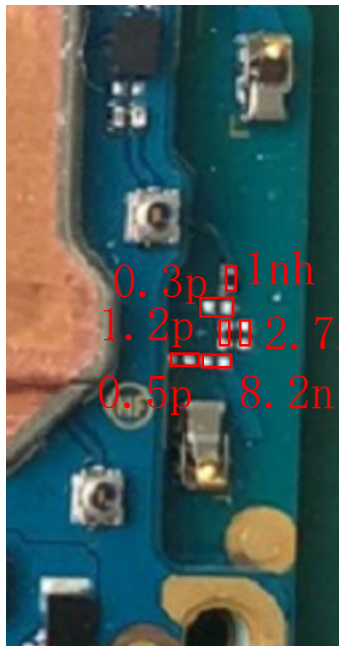
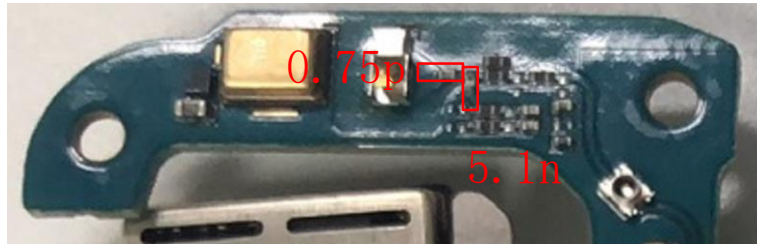
	SW3	SW4	SW2
Band	B12\17	B5	B71
value	39N	6.8n	7.5n

Main board spring sheet: All spring sheets need to avoid deformation/missing parts

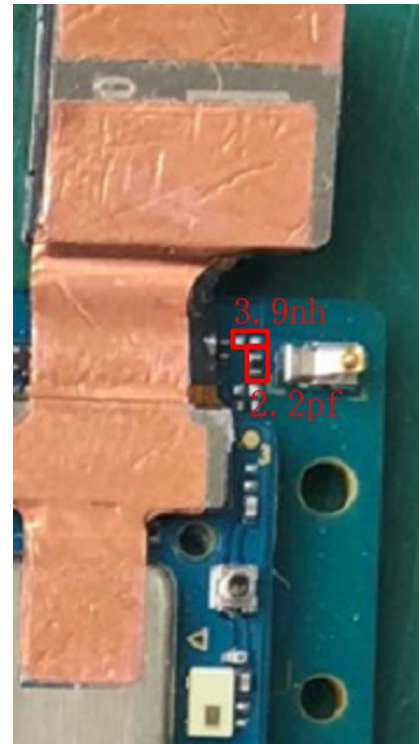


GPS L1/L5 antenna matching:

Wi-Fi ch1/GPSL1



TRX.N77



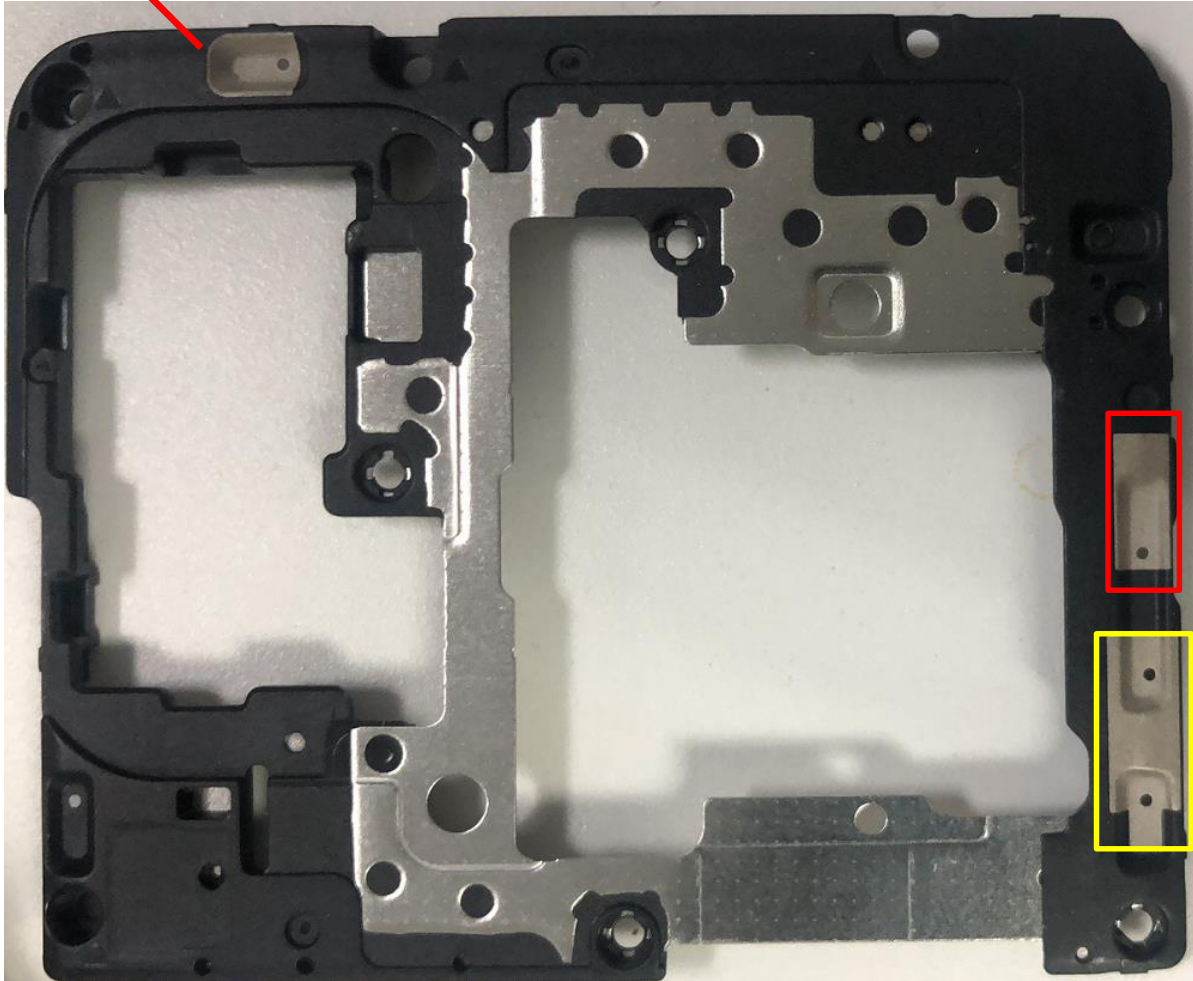
DRX.MH_N77



Wi-Fi ch0/GPSL5

Mid-frame LDS notice

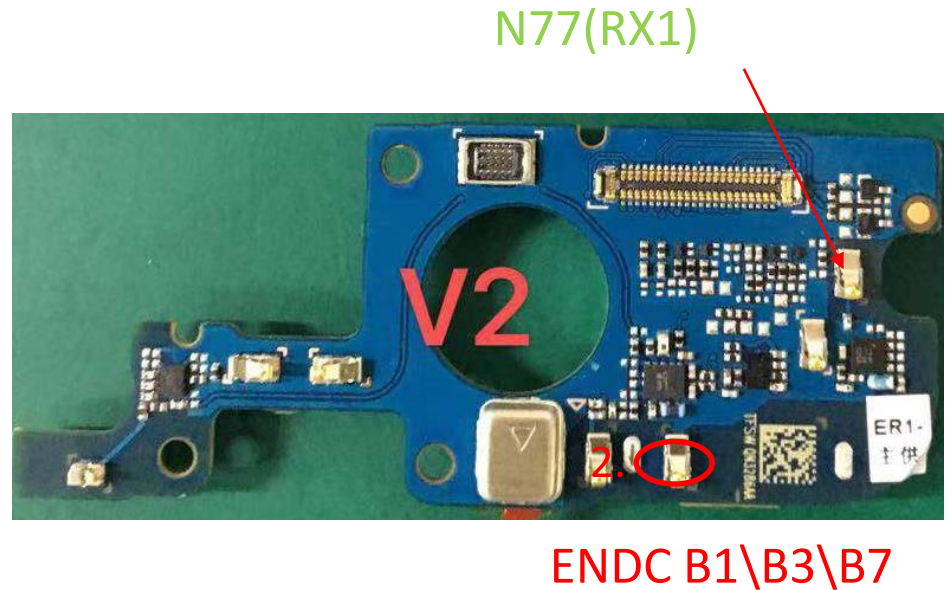
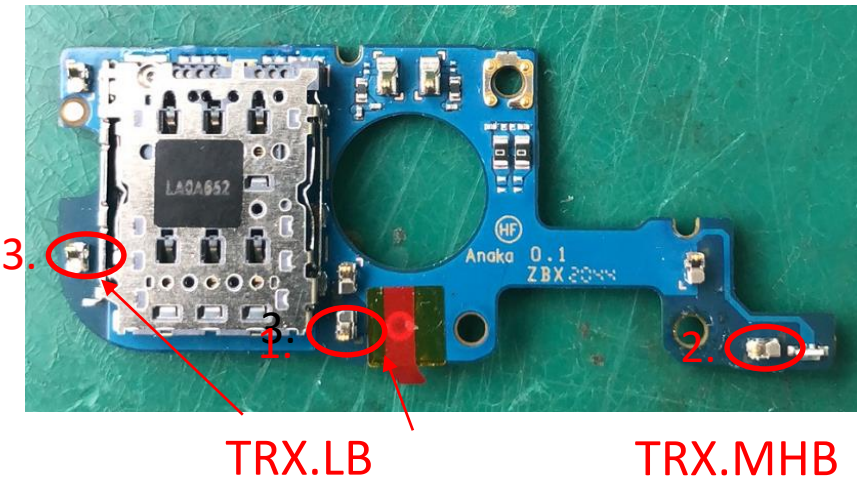
Ch1_WIFI6e



Ch0_WIFI6e

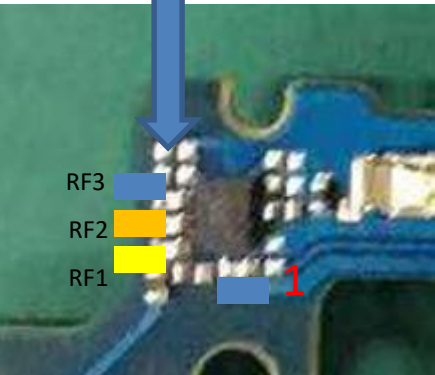
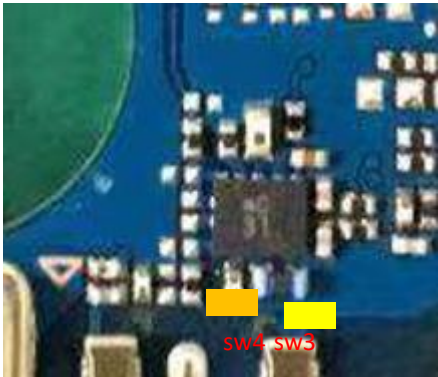
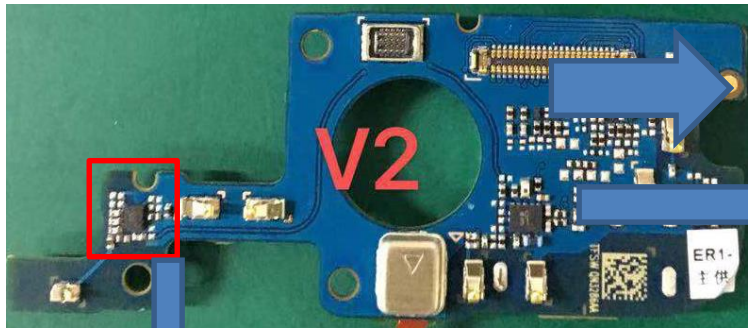
n77(TRX)

Main board spring sheet: All spring sheets need to avoid deformation/missing parts



Lower antenna notice:

1. Antenna aperture component location,



	SW3\4	SW4	SW3
Band	B2\5	B28	B71
value		18n	27n

	RF1	RF2	RF3
Band		MHb	
value	NC	3.3n	NC

Lower antenna notice:
BOX LDS antenna part

